

Lagrangians

ex:

$$\mathcal{L} = \frac{1}{2} m (\dot{x}^2 + \dot{y}^2)$$

switch to (r, θ) as coords

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$\dot{x} = \dot{r} \cos \theta - r \sin \theta \dot{\theta}$$

$$\dot{y} = \dot{r} \sin \theta + r \cos \theta \dot{\theta}$$

$$\dot{x}^2 + \dot{y}^2 = \dot{r}^2 + r^2 \dot{\theta}^2$$

$$\mathcal{L} = \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2)$$

$$\text{EL} \quad \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{x}} - \frac{\partial \mathcal{L}}{\partial x} = 0$$

$$\rightarrow \ddot{x} = 0$$

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{y}} - \frac{\partial \mathcal{L}}{\partial y} = 0$$

$$\rightarrow \ddot{y} = 0$$

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{r}} (r(t), \theta(t), \dot{r}(t), \dot{\theta}(t)) - \frac{\partial \mathcal{L}}{\partial r} (r(t), \theta(t), \dot{r}(t), \dot{\theta}(t)) \rightarrow$$

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\theta}} (r(t), \theta(t), \dot{r}(t), \dot{\theta}(t)) - \frac{\partial \mathcal{L}}{\partial \theta} (r(t), \theta(t), \dot{r}(t), \dot{\theta}(t)) \rightarrow$$

$$\frac{\partial \mathcal{L}}{\partial \dot{r}} = m \dot{r}$$

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}} = m r^2 \dot{\theta}$$

$$\frac{\partial \mathcal{L}}{\partial r} = m r \dot{\theta}^2$$

$$\frac{d}{dt} (m \dot{r}) - m r \dot{\theta}^2 = 0$$

$$\parallel \frac{d}{dt} (m r^2 \dot{\theta}) = 0$$

$\rightarrow$ Conservation of angular momentum



minimization problem

in calculus if $f: U \subseteq \mathbb{R} \rightarrow \mathbb{R}$, the min/max of f satisfy $f'(x) = 0$

generalize to functionals (functions of functions)

$L(\gamma)$ length of curve $\gamma: [0, 1] \rightarrow \mathbb{R}^2$

in Euclidean space $L(\gamma) = \int_0^1 \|\dot{\gamma}(t)\| dt$

$\nwarrow$ Lagrangian

$\downarrow$ action

Solve using variations

back to calculus

a variation of the point $x \in \mathbb{R}$ is a function $\varepsilon \in (-1, 1) \mapsto x_\varepsilon \in \mathbb{R}$

$$\text{s.t. } x_0 = x$$

$$f'(x) = 0 \Leftrightarrow \left. \frac{d}{d\varepsilon} f(x_\varepsilon) \right|_{\varepsilon=0} = 0 \quad \text{for all variations } x_\varepsilon$$

$$f'(x_0) \left. \frac{dx_\varepsilon}{d\varepsilon} \right|_{\varepsilon=0} = 0 \quad \text{for all variations}$$

$$\Leftrightarrow f'(x_0) = 0$$

generalize if S is a functional on some space of functions $\{\gamma: [t_0, t_1] \rightarrow M\}$

variation of γ : $(\varepsilon, t) \in (-1, 1) \times [t_0, t_1] \mapsto \gamma_\varepsilon(t) \in M$ (homotopy)

can also think of γ_ε as a curve for each ε

$$\text{require } \gamma_0 = \gamma.$$

γ is a critical point of S if $\left. \frac{d}{d\varepsilon} S(\gamma_\varepsilon) \right|_{\varepsilon=0} = 0$ for all variations of γ

(ex) $\gamma: [0, 1] \rightarrow \mathbb{R}$

$$S(\gamma) = \int_0^1 \gamma(t) dt$$

γ_ε arbitrary variation

$$\left. \frac{d}{d\varepsilon} S(\gamma_\varepsilon) \right|_{\varepsilon=0} = \int_0^1 \left. \frac{d}{d\varepsilon} \gamma_\varepsilon(t) \right|_{\varepsilon=0} dt \quad \text{not always 0}$$

no critical points

$$S(\gamma) = \int_0^1 \gamma(t)^2 dt$$

$$\left. \frac{d}{d\varepsilon} S(\gamma_\varepsilon) \right|_{\varepsilon=0} = \int_0^1 \left. \frac{d}{d\varepsilon} \gamma_\varepsilon^2(t) \right|_{\varepsilon=0} dt = \int_0^1 2\gamma(t) \left. \frac{d}{d\varepsilon} \gamma_\varepsilon(t) \right|_{\varepsilon=0} dt = 0 \quad \text{for all variations}$$

if $\gamma(t) = 0 \quad \forall t$

fix endpoints:

$$\gamma: [t_0, t_1] \rightarrow \mathbb{R}$$

$$\text{fix } \gamma(t_0) = a \quad \gamma(t_1) = b$$

variation with fixed endpoints: $\gamma_\varepsilon(t_0) = \gamma(t_0) \quad \gamma_\varepsilon(t_1) = \gamma(t_1) \quad \forall \varepsilon$

A special subclass of variational problems look like this:

$$S(\gamma) = \int_{t_0}^{t_1} \mathcal{L}(t, \gamma(t), \dot{\gamma}(t), \ddot{\gamma}(t), \dots) dt \quad \gamma: [t_0, t_1] \rightarrow \mathbb{R}^n$$

$$\mathcal{L}: \mathbb{R} \times \mathbb{R}^n \times \mathbb{R}^n \times \dots \times \mathbb{R}^n \rightarrow \mathbb{R} \quad (\text{Lagrangian})$$

in physics generally we stop at 1 derivative

Hamilton's principle: the physical path is a critical point of S .

Euler Lagrange eqs: $\left. \frac{d}{d\varepsilon} S(\gamma_\varepsilon) \right|_{\varepsilon=0} = 0 \quad \forall \text{ variations}$

$$\mathcal{L} = \mathcal{L}(t, q, \dot{q}) \quad q \in \mathbb{R}^n$$

$$S(\gamma_\varepsilon) = \int_{t_0}^{t_1} \mathcal{L}(t, \gamma_\varepsilon(t), \dot{\gamma}_\varepsilon(t)) dt$$

$$\left. \frac{d}{d\varepsilon} S(\gamma_\varepsilon) \right|_{\varepsilon=0} = \int_{t_0}^{t_1} \left. \frac{d}{d\varepsilon} \mathcal{L}(t, \gamma_\varepsilon(t), \dot{\gamma}_\varepsilon(t)) \right|_{\varepsilon=0} dt$$

$$= \int_{t_0}^{t_1} \left[\frac{\partial \mathcal{L}}{\partial q} (t, \gamma(t), \dot{\gamma}(t)) \left. \frac{d}{d\varepsilon} \gamma_\varepsilon(t) \right|_{\varepsilon=0} + \frac{\partial \mathcal{L}}{\partial \dot{q}} (t, \gamma(t), \dot{\gamma}(t)) \left. \frac{d}{d\varepsilon} \dot{\gamma}_\varepsilon(t) \right|_{\varepsilon=0} \right] dt$$

$$\left. \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}} \dot{\gamma}_\varepsilon(t) \right) \right|_{\varepsilon=0}$$

$$\left. \frac{d}{d\varepsilon} \gamma_\varepsilon(t) \right|_{\varepsilon=0} = \delta q(t)$$

integration by parts

$$= \int_{t_0}^{t_1} \left[\frac{\partial \mathcal{L}}{\partial q} (t, \gamma(t), \dot{\gamma}(t)) \delta q(t) - \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}} (t, \gamma(t), \dot{\gamma}(t)) \delta q(t) \right) + \left[\frac{\partial \mathcal{L}}{\partial \dot{q}} (t, \gamma(t), \dot{\gamma}(t)) \delta q(t) \right]_{t_0}^{t_1} \right] dt$$

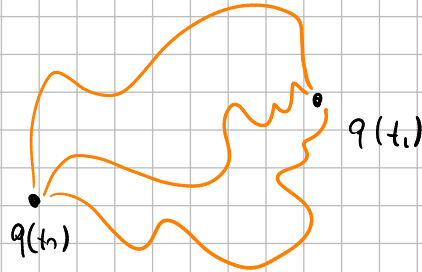
① γ is a critical point, then $\left. \frac{d}{d\varepsilon} S(\gamma_\varepsilon) \right|_{\varepsilon=0} = 0$ for all variations

$\Rightarrow$ so it's zero for all variations with fixed end points

$$\text{so } \left[\frac{\partial \mathcal{L}}{\partial \dot{q}} (t, \gamma(t), \dot{\gamma}(t)) \delta q(t) \right]_{t_0}^{t_1} = 0$$

$$\Rightarrow \int_{t_1}^{t_2} \left[\frac{\partial \mathcal{L}}{\partial q^n} (t, \gamma(t), \dot{\gamma}(t)) \delta q^n(t) - \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{s}^n} (t, \gamma(t), \dot{\gamma}(t)) \right) \delta q^n(t) \right] dt = 0 \quad \forall \delta q^n$$

$$\Rightarrow \boxed{\frac{\partial \mathcal{L}}{\partial q^n} (t, \gamma(t), \dot{\gamma}(t)) - \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{s}^n} (t, \gamma(t), \dot{\gamma}(t)) \right) = 0}$$



general form: $\mathcal{L} = \mathcal{L}(t, q_0, q_1, q_2, q_3, \dots, q_n)$

$$S = \int_{t_1}^{t_2} \mathcal{L}(t, \gamma(t), \dot{\gamma}(t), \ddot{\gamma}(t), \dots) dt$$

$$\underline{EL:} \quad \boxed{\sum_{k=0}^n (-1)^k \frac{d^k}{dt^k} \frac{\partial \mathcal{L}}{\partial q_k^n} (t, \gamma(t), \dot{\gamma}(t), \dots) = 0}$$

$$\underline{n=2} \quad \frac{\partial \mathcal{L}}{\partial q^0} - \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial q^1} + \frac{d^2}{dt^2} \frac{\partial \mathcal{L}}{\partial q^2} = 0$$

adding a total derivative to the Lagrangian does not change the result

often physicists write $\mathcal{L} = \frac{1}{2} m \dot{x}^2 - \frac{1}{2} k x^2$

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{x}} - \frac{\partial \mathcal{L}}{\partial x} = 0$$

proper way: $\mathcal{L} = \frac{1}{2} m \dot{s}^2 - \frac{1}{2} k x^2$

$$\mathcal{L}: \mathbb{R}^2 \rightarrow \mathbb{R}$$

$$(x, s) \mapsto \mathcal{L}(x, s)$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{s}} (x(t), \dot{x}(t)) \right) - \frac{\partial \mathcal{L}}{\partial x} (x(t), \dot{x}(t))$$

$$\frac{d}{dt} (m \dot{x}) + k x = 0 \quad \Rightarrow \quad m \ddot{x} = -k x$$

if $\mathcal{L} \Rightarrow \mathcal{L} + \frac{d}{dt}(f(x, \dot{x}))$ then same equations (what physicist say)

what they mean: if you add $F(x, \dot{x}, a)$ to $\mathcal{L}$

s.t. $\bar{F}(x(t), \dot{x}(t), \ddot{x}) = \frac{d}{dt}(f(x(t), \dot{x}(t)))$ then EL eq. are the same

$$\tilde{\mathcal{L}}(t, q, \dot{q}, a) = \mathcal{L}(t, q, \dot{q}) + \frac{\partial f}{\partial q} \dot{q} + \frac{\partial f}{\partial \dot{q}} a \quad \text{where } f = f(q, \dot{q})$$

$$\text{EL: } \frac{\partial \tilde{\mathcal{L}}}{\partial q}(t, x, \dot{x}, \ddot{x}) - \frac{d}{dt} \left(\frac{\partial \tilde{\mathcal{L}}}{\partial \dot{q}}(t, x, \dot{x}, \ddot{x}) \right) + \frac{d^2}{dt^2} \left(\frac{\partial \tilde{\mathcal{L}}}{\partial a}(t, x, \dot{x}, \ddot{x}) \right) = 0$$

$$\frac{\partial \mathcal{L}}{\partial q} + \cancel{\frac{\partial^2 f}{\partial q^2} \dot{x}} + \cancel{\frac{\partial^2 f}{\partial q \partial \dot{q}} \ddot{x}} - \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}} + \cancel{\frac{\partial^2 f}{\partial q \partial \dot{q}} \dot{q}} + \cancel{\frac{\partial f}{\partial q}} + \cancel{\frac{\partial^2 f}{\partial \dot{q}^2} a} \right) + \frac{d^2}{dt^2} \left(\cancel{\frac{\partial f}{\partial \dot{q}}} \right)$$

geometrically $\mathcal{L} = \mathcal{L}(q, \dot{q})$ $q \in M$ manifold
 $\dot{q} \in T_q M$

$$\mathcal{L}: TM \rightarrow \mathbb{R}$$

$$\ddot{x} = 0 \Leftrightarrow \begin{cases} \dot{\dot{q}} = 0 \\ \dot{x} = \dot{q} \end{cases}$$

$$M = \mathbb{R}^2$$

$$\mathcal{L}(q, \dot{q}) = \frac{1}{2} m \|\dot{q}\|^2$$

$$\dot{q} = \dot{x} \frac{\partial}{\partial x} + \dot{y} \frac{\partial}{\partial y}$$

$$\|\dot{q}\|^2 = \dot{x}^2 + \dot{y}^2$$

$$\left\langle \frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j} \right\rangle = \delta_{ij}$$

change coordinates:

r, θ

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$dx = dr \cos \theta - r \sin \theta d\theta$$

$$dy = dr \sin \theta + r \cos \theta d\theta$$

$$\frac{\partial}{\partial r} = \frac{\partial x}{\partial r} \frac{\partial}{\partial x} + \frac{\partial y}{\partial r} \frac{\partial}{\partial y}$$

$$\frac{\partial}{\partial \theta} = \frac{\partial x}{\partial \theta} \frac{\partial}{\partial x} + \frac{\partial y}{\partial \theta} \frac{\partial}{\partial y}$$

$$\mathbb{R}^4 = \mathbb{R} \times \mathbb{R}^3$$

path integral approach

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - m^2 \phi^2$$

$$S[\phi] = \int_{\mathbb{R}^4} d^4x \mathcal{L}(\phi, \partial\phi)$$

$\phi: \mathbb{R}^4 \rightarrow \mathbb{R}$

$$\mathcal{Z} = \int \mathcal{D}\phi e^{iS[\phi]}$$

translation invariant

2 problems:

① there is no Lebesgue measure on set of all fields

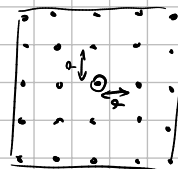
② exponential oscillates

to solve problem ① we can discretize spacetime

we lattice $\Lambda = \left\{ \sum_n \alpha^n e_n, \alpha^n = -N, -N+1, \dots, N-1, N \right\} \quad a = \frac{L}{N}$

$$= \{x_i\}_{i \in \mathbb{I}}$$

$$\hookrightarrow \mathcal{Z} = \prod_i \int_{\mathbb{R}} d\phi_i e^{iS[\phi]}$$



② "solve" it by doing a Wick rotation: $t \rightarrow it$

$$i \int_{\mathbb{R}^4} d^4x \frac{1}{2} \left((\partial_t \phi)^2 - (\nabla \phi)^2 \right)$$

$$dt \rightarrow i dt$$

$$\frac{\partial}{\partial t} \rightarrow -i \frac{\partial}{\partial t}$$

$$\rightarrow i^2 \int_{\mathbb{R}^4} d^4x \frac{1}{2} \left(-(\partial_t \phi)^2 - (\nabla \phi)^2 \right)$$

$$= \frac{1}{2} \int_{\mathbb{R}^4} d^4x \left((\partial_t \phi)^2 + (\nabla \phi)^2 \right)$$

$\ll$ integration by parts

$$= -\frac{1}{2} \int_{\mathbb{R}^4} d^4x \left(\phi \partial_t^2 \phi + \phi \nabla^2 \phi \right)$$

$$\nabla \phi \cdot \nabla \phi = \nabla \cdot (\phi \nabla \phi) - \phi \nabla^2 \phi$$

$$e^{-\frac{1}{2} \int_{\mathbb{R}^4} d^4x (\phi L \phi)}$$

$L = \partial_t^2 + \nabla^2$ linear differential operator

→ probability measure

mult. -dimensional Gaussian on $\mathbb{R}^n$

$$e^{-\frac{1}{2} x^t A x}$$

A matrix (symmetric + pos. definite)

Gaussian distribution (1D)

$$p(x) = e^{-\frac{1}{2} x^2}$$

$$\langle f \rangle = \frac{\int_{-\infty}^{\infty} dx e^{-\frac{1}{2} x^2} f(x)}{\int_{-\infty}^{\infty} dx e^{-\frac{1}{2} x^2}}$$

nD version

$$p(\vec{x}) = e^{-\frac{1}{2} x^t A x}$$

$$\langle x^i \rangle = \frac{\int_{\mathbb{R}^n} d^n x e^{-\frac{1}{2} x^t A x} x^i}{\int_{\mathbb{R}^n} d^n x e^{-\frac{1}{2} x^t A x}}$$

$$\langle \phi(x) \rangle = \frac{\int D\phi e^{S[\phi]} \phi(x)}{\int D\phi e^{S[\phi]}} = 0 \quad (1\text{-point function})$$

$\downarrow$
 $x \in \mathbb{R}^4$

quantization step

(interpretation of $\langle \phi(x^1) \phi(x^2) \phi(x^3) \dots \rangle$)

$$\langle 0 | \hat{\phi}(t, \vec{x}) | 0 \rangle$$

Theorem: Isserlis theorem

For a gaussian distribution $p(x) = e^{-\frac{1}{2} x^t A x}$ $x \in \mathbb{R}^n$

$$\langle x^{i_1} x^{i_2} \dots x^{i_k} \rangle = \frac{\int_{\mathbb{R}^n} d^n x p(x) x^{i_1} \dots x^{i_k}}{\int_{\mathbb{R}^n} d^n x p(x)}$$

$$= \sum_{\sigma \in P_k^2} \prod_{\{r,s\} \in \sigma} \langle x^{ir} x^{is} \rangle$$

P_k^2 set of all partitions into pairs of $\{1, \dots, k\}$

Corollary: $= 0$ if k is odd

ex: $k=4$ $\{1, 2, 3, 4\}$

partitions:
 $(1,2) (3,4)$
 $(1,3) (2,4)$
 $(1,4) (2,3)$

$$\langle x^{i1} \dots x^{i4} \rangle = \langle x^{i1} x^{i2} \rangle \langle x^{i3} x^{i4} \rangle + \langle x^{i1} x^{i3} \rangle \langle x^{i2} x^{i4} \rangle + \langle x^{i1} x^{i4} \rangle \langle x^{i2} x^{i3} \rangle$$

for path integral:

$$\langle \phi(x^{i1}) \dots \phi(x^{i4}) \rangle = \sum_{\sigma \in P_k^2} \prod_{\{r,s\} \in \sigma} \langle \phi(x^{ir}) \phi(x^{is}) \rangle$$

figure out the 2-point function

$$\langle 0 | \hat{\phi}(x) \hat{\phi}(x') | 0 \rangle$$

for a Gaussian

$$\langle x^i x^j \rangle = \frac{\int_{\mathbb{R}^n} d^n x e^{-\frac{1}{2} x^t A x} x^i x^j}{\int_{\mathbb{R}^n} d^n x e^{-\frac{1}{2} x^t A x}} = (A^{-1})_{ij}$$

for path integrals

$$\langle \phi(x) \phi(x') \rangle := (L^{-1})(x, x')$$

$$P(\phi) = e^{-\frac{1}{2} \int_{\mathbb{R}^4} d^4 x (\phi L \phi)}$$

$$L^{-1}(x, x') = G(x, x')$$

s.t.

$$L G(x, x') = \delta(x - x')$$

Green's function

in our case

$$L = \partial_t^2 + \nabla^2$$

$$L^{-1} = \frac{1}{\partial_t^2 + \nabla^2}$$

FT

$$(k_0^2 + \vec{k}^2) \hat{G}(k) = 1$$

Feynman theorem:

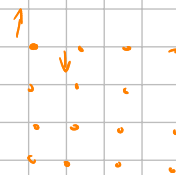
$$\frac{1}{k_0^2 + \vec{k}^2}$$

in QM are the same

$$Z[q] = \int Dq e^{\frac{i}{\hbar} S[q]} \quad \text{"sum over all paths"}$$

Statistical mechanics

$$Z = \sum_i e^{-\beta E_i}$$



Connection (path integral $\leftrightarrow$ canonical quantization)

$$\langle \phi(x_1) \phi(x_2) \dots \phi(x_n) \rangle \equiv \langle 0 | T(\hat{\phi}(x_1) \dots \hat{\phi}(x_n)) | 0 \rangle$$

$\epsilon \in \mathbb{R}^4$ $\downarrow$ time ordering

$$[\hat{\phi}(t, \vec{x}), \hat{\phi}(t, \vec{y})] = 0$$

$$[\hat{q}_i, \hat{q}_j] = 0$$

$$[\hat{\phi}(t, \vec{x}), \phi(t', \vec{x}')] = 0 \quad \text{if } (t, \vec{x}), (t', \vec{x}') \text{ are spacelike separated}$$

interactions

$$S[\phi] = \underbrace{\int_{\mathbb{R}^4} d^4x \mathcal{L}_{\text{free}}(\phi, \partial\phi)}_{S_0[\phi]} + g \int_{\mathbb{R}^4} d^4x \mathcal{L}_{\text{int}}(\phi, \partial\phi) = S_0[\phi] + g S_I[\phi]$$

trick

$$Z[\phi] = \int D\phi e^{iS_0[\phi] + igS_I[\phi]} = \sum_{k=0}^{\infty} \frac{(ig)^k}{k!} \int D\phi e^{iS_0[\phi]} (S_I(\phi))^k$$

$\int d^3x (\alpha_3 \phi^3(x) + \alpha_4 \phi^4(x) + \dots)$

$$\langle \phi(x) \phi(y) \rangle = \sum_{k=0}^{\infty} \frac{(ig)^k}{k!} \int D\phi e^{iS_0[\phi]} (S_I(\phi))^k \phi(x) \phi(y)$$

$$\sum_{k=0}^{\infty} \frac{(ig)^k}{k!} \int D\phi e^{iS_0[\phi]} (S_I(\phi))^k$$

$$o(1) : \langle \phi(x) \phi(y) \rangle_0 \leftarrow \text{free}$$

$$x \text{ --- } y$$

$$o(g) : \langle \phi(x) \phi(y) \rangle_0$$

$$x \text{ --- } y$$

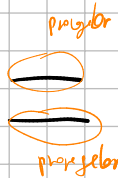
$$o(g^2) :$$

$$\text{---} + g \rightarrow \text{---} \circ \text{---}$$

$$\boxed{\text{---} \circ \text{---}}$$

$$\text{---} \circ \text{---}$$

3-point



$$\langle \phi(x) \phi(y) \phi(z) \rangle = \frac{\sum_{k=0}^{\infty} \frac{(ig)^k}{k!} \int D\phi e^{iS_0[\phi]} (S_I(\phi))^k \phi(x) \phi(y)}{\sum_{k=0}^{\infty} \frac{(ig)^k}{k!} \int D\phi e^{iS_0[\phi]} (S_I(\phi))^k}$$

expd to $o(g)$

$$= \langle \phi(x) \phi(y) \phi(z) \rangle_0 + \boxed{ig \int d^4(s) \langle \phi(x) \phi(y) \phi(z) \phi^3(s) \rangle_0}$$

$$- ig \langle \phi(x) \phi(y) \phi(z) \rangle_0 \int d^4(s) \langle \phi^3(s) \rangle_0$$

something wrong here

$$1 \ 2 \ 3 \ 4 \ 5 \ 6 =$$

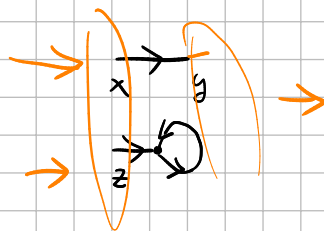
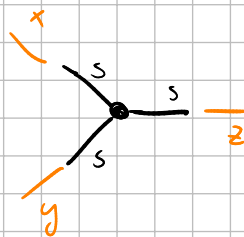
$$(12)(34)(56)$$

$$(12)(35)(46)$$

$$(13)(24)(56)$$

$$(13)(25)(46)$$

$$(14)(23)$$



$$\langle \phi(x) \phi(y) \rangle \langle \phi(z) \phi(s) \rangle \langle \phi(s)^2 \rangle$$

$$\langle \phi(x) \phi(z) \rangle \langle \phi(y) \phi(s) \rangle \langle \phi(s)^2 \rangle$$

$$\langle \phi(y) \phi(z) \rangle \langle \phi(x) \phi(s) \rangle \langle \phi(s)^2 \rangle$$

$$\langle \phi(x) \phi(s) \rangle \langle \phi(y) \phi(s) \rangle \langle \phi(z) \phi(s) \rangle$$

$$\langle \phi(x) \rangle = -\phi$$

momentum space

$$\phi(p) = \int \frac{d^4 x}{(2\pi)^4} \phi(x) e^{-ip \cdot x}$$

$$\phi(x) = \int \frac{d^4 p}{(2\pi)^4} \phi(p) e^{ip \cdot x}$$

$$S[\phi] = -\frac{1}{2} \int d^4 x \phi \nabla^2 \phi + g \int d^4 x \phi^3(x)$$

$$= \frac{1}{2} \int d^4 x \int d^4 p_1 \int d^4 p_2 \frac{1}{(2\pi)^4} \phi(p_1) \phi(p_2) p_2^2 e^{i(p_1+p_2) \cdot x} = (2\pi)^4 \delta(p_1+p_2)$$

$$+ g \int d^4 x \int d^4 p_1 \int d^4 p_2 \int d^4 p_3 \frac{1}{(2\pi)^6} \phi(p_1) \phi(p_2) \phi(p_3) e^{i(p_1+p_2+p_3) \cdot x} = (2\pi)^4 \delta(p_1+p_2+p_3)$$

$$= \frac{1}{2} \int d^4 p (-p^2 \phi(p) \phi(-p)) + \frac{g}{(2\pi)^2} \int d^4 p_1 \int d^4 p_2 \int d^4 p_3 \phi(p_1) \phi(p_2) \phi(p_3) \delta(p_1+p_2+p_3)$$

$$\langle \phi(p_1) \dots \phi(p_n) \rangle = \frac{\sum_{k=0}^{\infty} g^k \int D\phi e^{-S_0[\phi]} S_I(\bar{I}\phi) \phi(p_1) \dots \phi(p_n)}{Z}$$

$$\langle \phi(p_1) \phi(p_2) \rangle_0 = \frac{1}{(p_1+p_2)^2} S_{12}$$

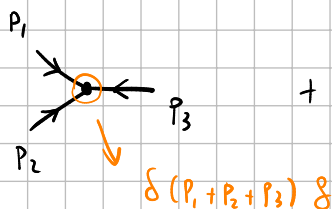
$$S_{ij} = (p_i + p_j)^2$$

$$\frac{1}{2} \int d^4 p (-p^2 \phi(p) \phi(-p))$$

$$\downarrow$$

$$\frac{1}{2} \int d^4 p_1 \int d^4 p_2 (-p_1 \cdot p_2 \phi(p_1) \phi(p_2)) \delta(p_1+p_2)$$

$$\langle \phi(p_1) \phi(p_2) \phi(p_3) \rangle = \text{diagram} + O(g^2)$$



$$\langle \varphi(p_1) \varphi(p_2) \varphi(p_3) \varphi(p_4) \rangle =$$

sum over permutations of only non-ghost mes

$$2 \left(\frac{1}{S_{23}} + \frac{1}{S_{34}} \right) + \left(\frac{1}{S_{12}} + \frac{1}{S_{14}} \right) + \left(\frac{1}{S_{12}} + \frac{1}{S_{13}} \right) + \left(\frac{1}{S_{24}} + \frac{1}{S_{14}} \right) + \left(\frac{1}{S_{34}} + \frac{1}{S_{14}} \right)$$

$$2 \left(\frac{1}{12} + \frac{1}{14} \right) + \left(\frac{1}{12} + \frac{1}{13} \right) + \left(\frac{1}{14} + \frac{1}{13} \right)$$

$$8 \frac{1}{S_{12}} + 8 \frac{1}{S_{14}} + 4 \frac{1}{S_{13}}$$

eq 2

$$\mathcal{A}_n = \sum_{\sigma \in S_n / \mathbb{Z}_n} \mathcal{A}_n(\sigma(1), \dots, \sigma(n))$$

ordered

$$\mathcal{A}_4(1, 2, 3, 4) = \frac{1}{S_{12}} + \frac{1}{S_{14}}$$

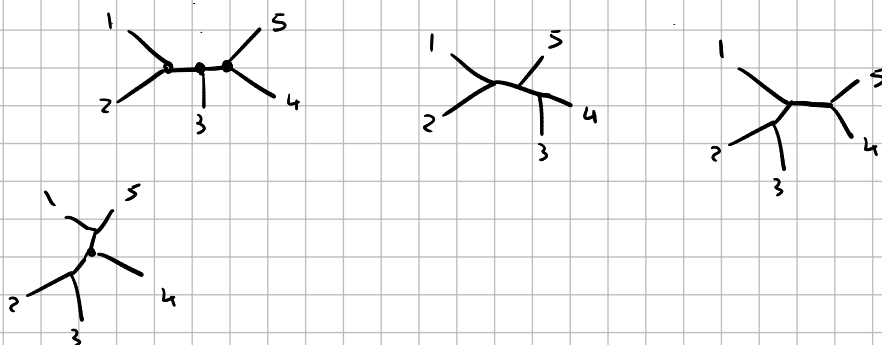
$$1234 \sim 2341$$

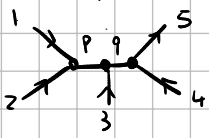
✓ 1 2 3 4	3 1 2 4
1 2 4 3	3 1 4 2
1 3 4 2	3 2 1 4
1 3 2 4	3 2 4 1
1 4 2 3	3 4 1 2
1 4 3 2	3 4 2 1
2 1 3 4	4 1 2 3
2 1 4 3	4 1 3 2
2 3 1 4	4 2 1 3
2 3 4 1	4 2 3 1
2 4 1 3	4 3 1 2
2 4 2 1	4 3 2 1

$$4 \left(\frac{1}{S_{12}} + \frac{1}{S_{13}} + \frac{1}{S_{14}} \right)$$

12	14
12	13
13	12
13	14
14	13
14	12

5 point



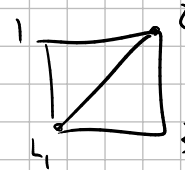
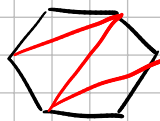
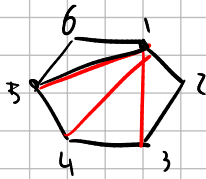
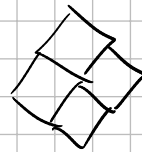
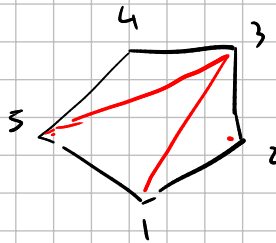
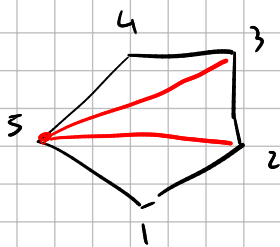


$$\int d^3 q \int d^3 p \frac{1}{p^2} \delta(p_1 + p_2 - p) \delta(p - q + p_3) \frac{1}{q^2} \delta(p + p_3 - q) \delta(q + p_5 + p_4)$$

$$\frac{1}{(p_1 + p_2)^2} \frac{1}{(p_4 + p_5)^2} = \frac{\delta(p_1 + p_2 + p_3 + p_4)}{s_{12} s_{45}}$$

$$\frac{1}{s_{12} s_{45}} + \frac{1}{s_{12} s_{34}} + \frac{1}{s_{23} s_{45}} + \frac{1}{s_{23} s_{15}} + \frac{1}{s_{34} s_{15}}$$

$$s_{ij} = (p_i + p_j)^2 = 2 p_i \cdot p_j$$



$$\frac{\partial}{\partial r} = \cos\theta \frac{\partial}{\partial x} + \sin\theta \frac{\partial}{\partial y}$$

$$\vec{v} = \vec{v}_r \frac{\partial}{\partial r} + \vec{v}_\theta \frac{\partial}{\partial \theta}$$

$$\frac{\partial}{\partial \theta} = -r \sin\theta \frac{\partial}{\partial x} + r \cos\theta \frac{\partial}{\partial y}$$

$$\|\vec{v}\|^2 = ?$$

$$\left\langle \frac{\partial}{\partial r}, \frac{\partial}{\partial r} \right\rangle = \left\langle \cos\theta \frac{\partial}{\partial x} + \sin\theta \frac{\partial}{\partial y}, \cos\theta \frac{\partial}{\partial x} + \sin\theta \frac{\partial}{\partial y} \right\rangle$$

$$= (\cos\theta)^2 + (\sin\theta)^2 = 1$$

$$\left\langle \frac{\partial}{\partial r}, \frac{\partial}{\partial \theta} \right\rangle = \left\langle \cos\theta \frac{\partial}{\partial x} + \sin\theta \frac{\partial}{\partial y}, -r \sin\theta \frac{\partial}{\partial x} + r \cos\theta \frac{\partial}{\partial y} \right\rangle$$

$$= -r \cos\theta \sin\theta + r \cos\theta \sin\theta = 0$$

$$\left\langle \frac{\partial}{\partial \theta}, \frac{\partial}{\partial \theta} \right\rangle = \left\langle -r \sin\theta \frac{\partial}{\partial x} + r \cos\theta \frac{\partial}{\partial y}, -r \sin\theta \frac{\partial}{\partial x} + r \cos\theta \frac{\partial}{\partial y} \right\rangle$$

$$= r^2$$

Physics in this

$$\mathcal{L} = \frac{1}{2} m (\dot{x}^2 + \dot{y}^2)$$

$$\begin{aligned} x &= r \cos\theta \\ y &= r \sin\theta \end{aligned}$$

$$\dot{x} = \dot{r} \cos\theta - r \sin\theta \dot{\theta}$$

$$\dots \rightarrow \mathcal{L} = \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2)$$

Mechanics

q position of a particle
 $\vec{v}$ velocity of particle

$V(q)$ potential energy

$$\mathcal{L} = \frac{1}{2} m \vec{v}^2 - V(q) \quad \rightarrow \quad EL: \quad m \vec{v} = - \nabla V(q) = \text{force}$$

for fields

$$\mathcal{L} = \frac{1}{2} m \left(\frac{d}{dt} x \right)^2 - V(x)$$

$$\parallel \quad x = x(t)$$

↓ field

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - V(\phi)$$

$$\parallel \quad \phi = \phi(t, x)$$

$$\boxed{\frac{\partial}{\partial x^\mu} \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} - \frac{\partial \mathcal{L}}{\partial \phi} = 0}$$

$$\mathcal{L} = \frac{1}{2} \dot{\phi}^2 - V(\phi)$$

$$\frac{\partial}{\partial x^\mu} \frac{\partial \mathcal{L}}{\partial \dot{\phi}} (\phi, \dot{\phi}) - \frac{\partial \mathcal{L}}{\partial \phi} (\phi, \dot{\phi})$$

in Minkowski: use $\partial_\mu^2 = \partial_\mu^2$

$$\rightarrow \partial_\mu \partial^\mu \phi = -V'(\phi)$$

in QFT

start with free Lagrangian for your kind of field

$$\mathcal{L}_F = \frac{1}{2} (\partial_\mu \phi \partial^\mu \phi - m^2 \phi^2)$$

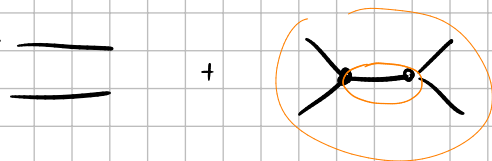
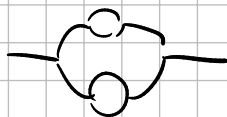
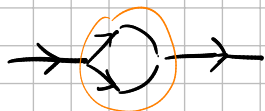
$$\mathcal{L}_F = -\frac{1}{4} \text{tr}(F_{\mu\nu} F^{\mu\nu})$$

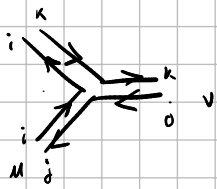
ex: add ϕ^3 interaction: $\mathcal{L} = \frac{1}{2} (\partial_\mu \phi \partial^\mu \phi - m^2 \phi^2) + g \phi^3$

canonical quantization

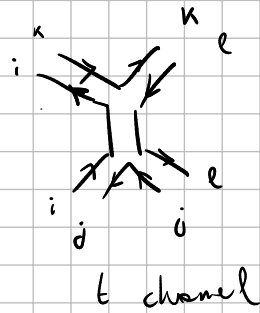
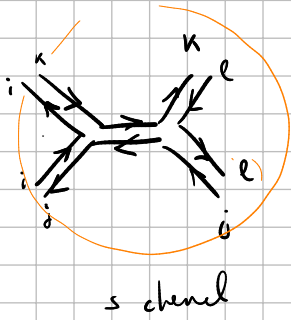
$$\phi(p) \rightarrow \hat{\phi}(p) \quad (\text{operator})$$

$$\langle 0 | \phi(q) \phi(p) | 0 \rangle = \delta(p-q) \frac{1}{p^2} \quad (\text{free Lagrangian})$$





$$(A_m)^i_j$$

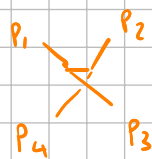


planar graphs

$$A_n = \sum_{\sigma \in S_n / \mathbb{Z}_n} \left(\text{diagram} \right) A_n(\sigma(1), \sigma(2), \dots, \sigma(n))$$

$$A_4(1, 2, 3, 4) = \text{diagram 1} + \text{diagram 2} \rightarrow \frac{1}{(p_1 + p_2)^2}$$

$$A_4(1, 2, 4, 3) = \text{diagram 3} + \text{diagram 4} \rightarrow \frac{1}{(p_1 + p_2)^2}$$



///

$$A_4(2, 1, 3, 4) = \text{diagram 5} + \text{diagram 6}$$

• tree-level diagrams: no loops



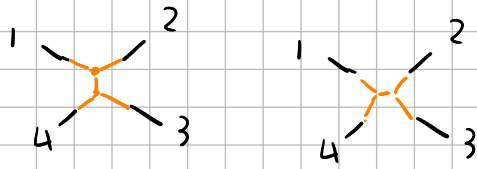
not allowed!

• fixed ordering

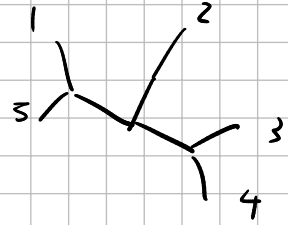
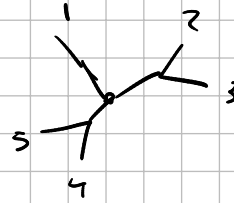
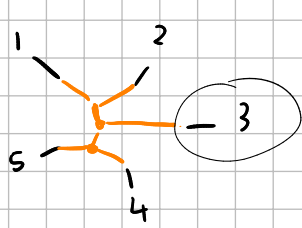
• planar diagrams

• connected

$$n = 4$$

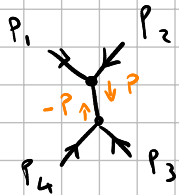


$$n = 5$$

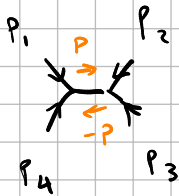


rule each internal edge is a propagator $\frac{1}{p^2}$

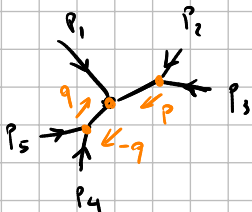
at each vertex we have conservation of momentum



$$\frac{1}{p^2} = \frac{1}{(p_1 + p_2)^2} = \frac{1}{(-p_3 - p_4)^2}$$

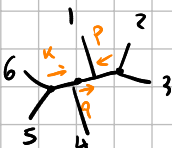
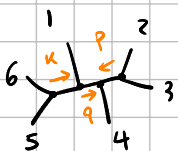


$$\frac{1}{p^2} = \frac{1}{(p_1 + p_4)^2} = \frac{1}{(p_2 + p_3)^2}$$



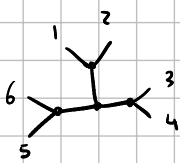
$$\frac{1}{(p_2 + p_3)^2} \quad \frac{1}{(p_4 + p_5)^2}$$

$$-q = p_1 + p = p_1 + p_2 + p_3 = -p_4 - p_5$$



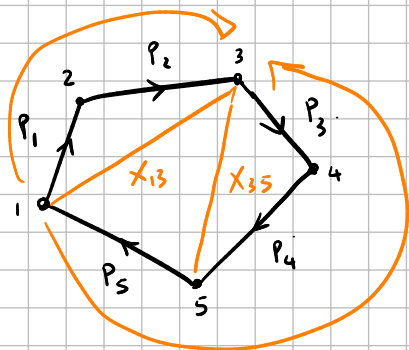
$$\frac{1}{(p_2 + p_3)^2} \cdot \frac{1}{(p_2 + p_3 + p_4)^2} \cdot \frac{1}{(p_5 + p_6)^2}$$

$X_{24} \quad X_{25} \quad X_{51}$



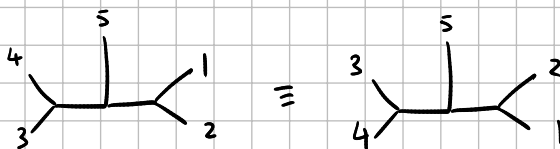
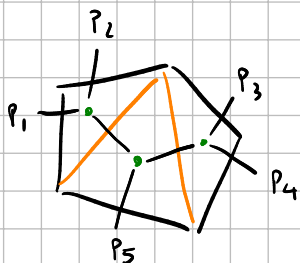
$$\frac{1}{(p_1 + p_2)^2} \cdot \frac{1}{(p_3 + p_4)^2} \cdot \frac{1}{(p_5 + p_6)^2}$$

$x_{1,3} \quad x_{3,5} \quad x_{5,1}$

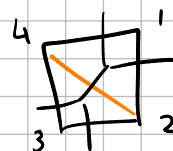
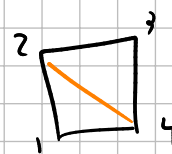
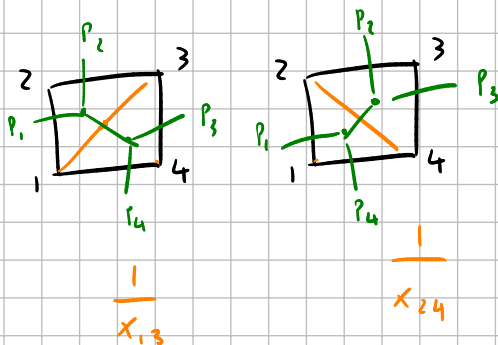


$$\sum_i p_i = 0$$

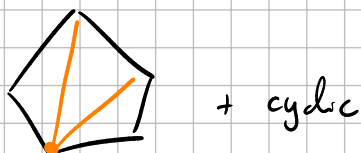
$$\frac{1}{x_{1,3}} \frac{1}{x_{3,5}} = \frac{1}{(p_1 + p_2)^2} \frac{1}{(p_3 + p_4)^2}$$



$n = 4$

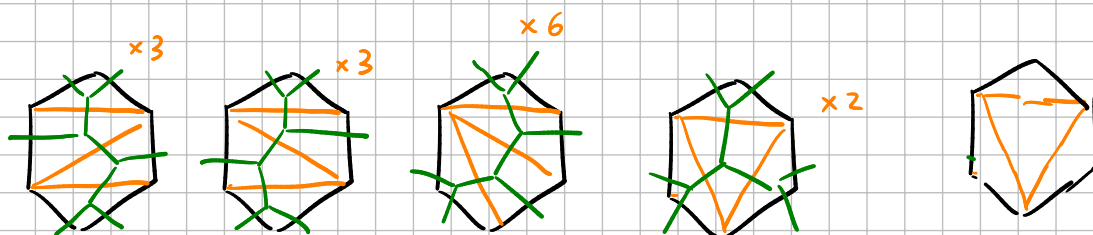


$n = 5$



+ cyclic

$n = 6$



number of planar diagrams with fixed ordering is the Catalan number

C_n

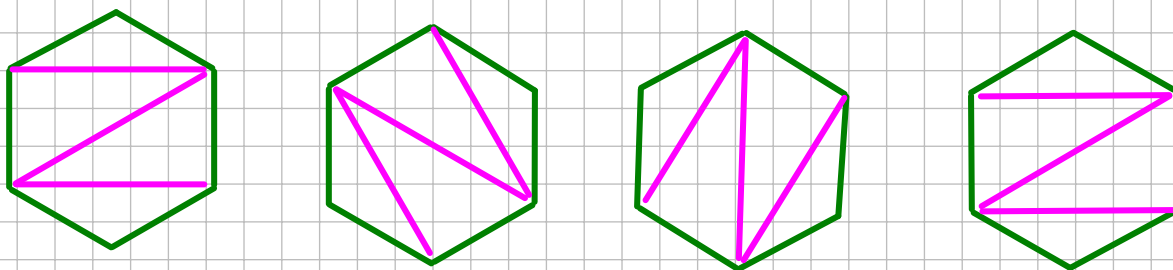
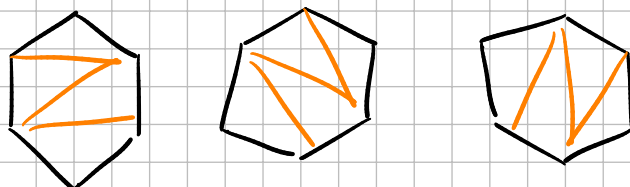
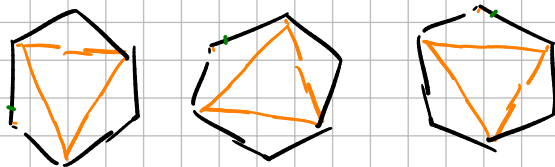
$C_0 = 1$

$C_1 = 1$

$C_2 = 2$

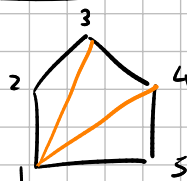
$C_3 = 5$

$C_4 = 14$



$$\frac{1}{n+1} \binom{2n}{n}$$

triangulation τ



pick $X_{13} X_{14}$ as planar

everything but $c_{i-1, j-1}$ with $(i, j) \in \{(1, 3), (1, 4)\}$

exclude $c_{52} c_{53}$ and

$c_{15} c_{45}$

$\Rightarrow$ because planar

and $c_{12} c_{23} c_{34}$

c_{12}	c_{13}	c_{14}	c_{15}
	c_{23}	c_{24}	c_{25}
		c_{34}	c_{35}
			c_{45}

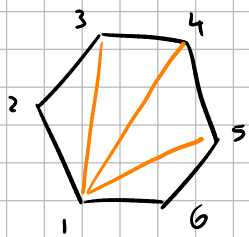
$$c_{12} = -X_{13} = -(P_1 + P_2)^2$$

$n = 5$

number of inversions: $\frac{n(n-3)}{2} = 5$

planar: sum of consecutive momenta squared

$$c_{ij} = -(P_i + P_j)^2$$



take

$X_{13} \ X_{14} \ X_{15}$

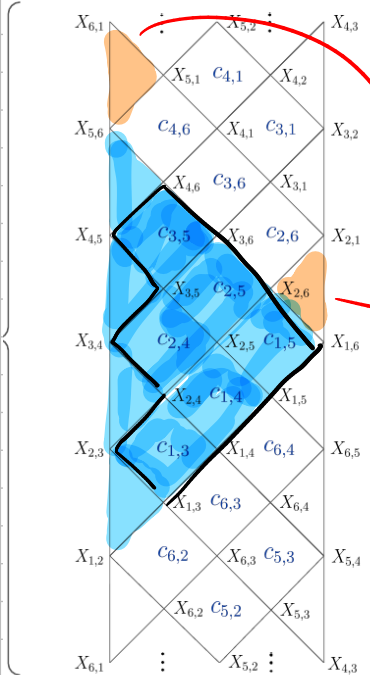
exclude

$C_{62} \ C_{63} \ C_{64}$

$C_{i-1, i-1}$

C_{12}	C_{13}	C_{14}	C_{15}	C_{16}
	C_{23}	C_{24}	C_{25}	C_{26}
		C_{34}	C_{35}	C_{36}
			C_{45}	C_{46}
				C_{56}

$$C_{ij} = -(P_i + P_j)^2$$



planar

$$C_{16} = -X_{62}$$

replace

X_{24}	with	C_{13}
X_{25}	with	C_{24}
X_{26}	with	C_{15}
X_{35}		C_{24}
X_{36}		C_{25}
X_{46}		C_{35}

$n=6$:

3 independent variables

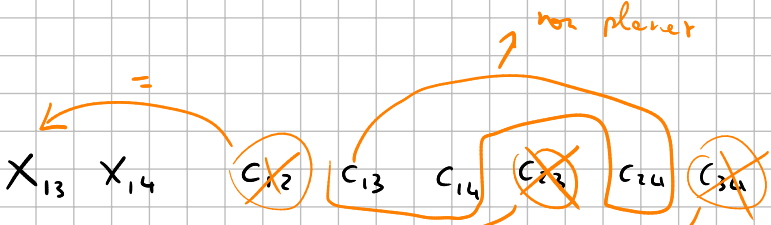
in the $n=5$ example:

want to show that

the remaining ones

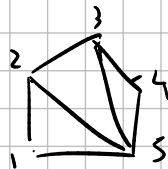
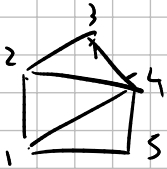
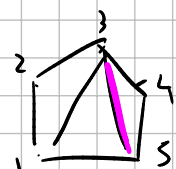
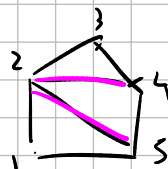
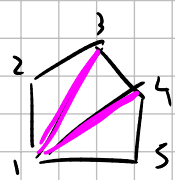
$(X_{24} \ X_{25} \ X_{35})$

are obtained from basis



$X_{13} \ X_{14}$

$$X_{12} = P_1^2 = 0$$



→ these pictures are to find the non-zero X_{ij} s

$$C_{ij} = X_{ij} + X_{i+1, j+1} - X_{i, j+1} - X_{i+1, j}$$

$$C_{13} = X_{13} + X_{24} - X_{14}$$

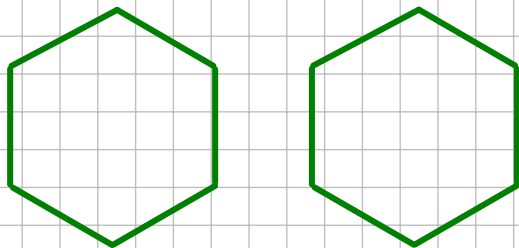
$$C_{14} = X_{14} + X_{25} - X_{24}$$

$$C_{24} = X_{24} + X_{35} - X_{25}$$

$$X_{25} = C_{14} + C_{13} - X_{13}$$

$$X_{35} = C_{14} + C_{24} - X_{14}$$

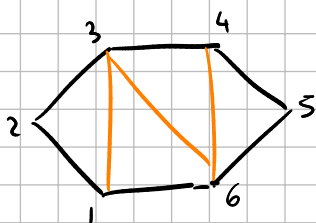
$$X_{24} = C_{24} - X_{35} + X_{25}$$



Procedure to replace variables:

• choose triangulation, keep the chords

• for each discarded X_{ij} , replace with $C_{i-1, j-1}$



replace:

$$X_{13} \quad X_{36} \quad X_{46}$$

$$X_{14} \rightarrow C_{63}$$

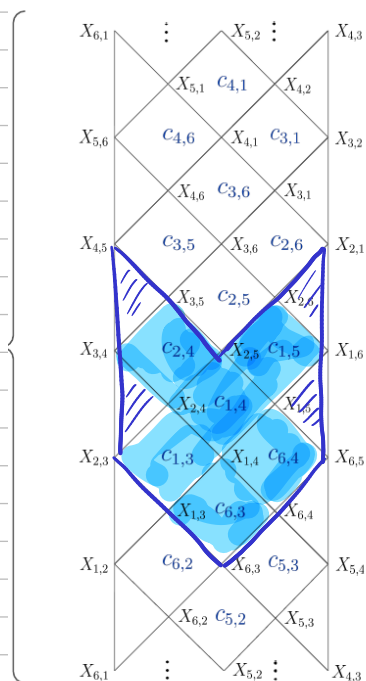
$$X_{15} \rightarrow C_{64}$$

$$X_{24} \rightarrow C_{13}$$

$$X_{25} \rightarrow C_{14}$$

$$X_{26} \rightarrow C_{15}$$

$$X_{35} \rightarrow C_{24}$$

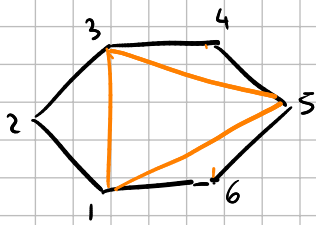


$$C_{23} = X_{23} + X_{34} - X_{24} - X_{33} \\ \parallel \\ - X_{24}$$

$$\text{if } i \leq j \quad X_{ij} = \left(\sum_{k=i}^{j-1} p_k \right)^2$$

$$X_{i,i+1} = p_i^2 = 0$$

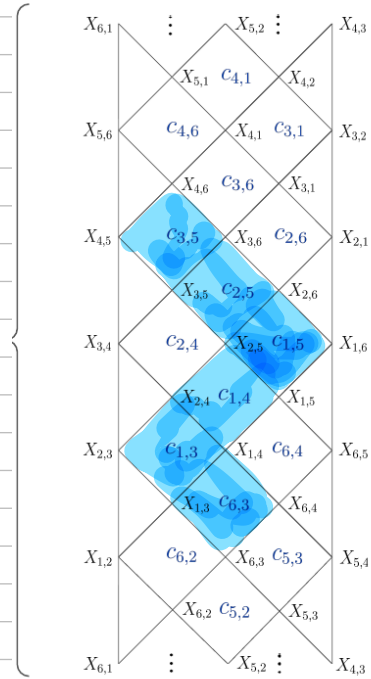
$$X_{i,i} = 0 \quad \text{by definition}$$



$$X_{13} \quad X_{35} \quad X_{51}$$

replace:

$$\begin{aligned}
 X_{14} &\rightarrow C_{63} \\
 X_{24} &\rightarrow C_{13} \\
 X_{25} &\rightarrow C_{14} \\
 X_{26} &\rightarrow C_{15} \\
 X_{36} &\rightarrow C_{25} \\
 X_{46} &\rightarrow C_{35}
 \end{aligned}$$



$$\mathcal{L} = \frac{1}{2} \text{tr}(\partial_\mu \phi \partial^\mu \phi) + g \text{tr}(\phi^3)$$

ϕ is an $N \times N$ matrix

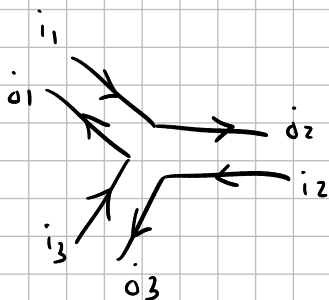
ϕ^i_j ← color indices

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi^i_j \partial^\mu \phi^j_i + g \phi^i_j \phi^j_k \phi^k_i$$

$$\langle \phi^i_j(x) \phi^k_l(y) \rangle = \frac{\int D\phi e^{iS[\phi]} \phi^i_j(x) \phi^k_l(y)}{\int D\phi e^{iS[\phi]}}$$

$$\langle \phi^{i_1}_{j_1}(x_1) \phi^{i_2}_{j_2}(x_2) \rangle_0 \propto \delta^{i_1}_{j_2} \delta^{i_2}_{j_1}$$

free



$$\langle \phi^{i_1}_{j_1}(p_1) \phi^{i_2}_{j_2}(p_2) \phi^{i_3}_{j_3}(p_3) \rangle$$

$$e^{S[\phi]} = e^{S_0[\phi]} \left(1 + g \int \text{tr}(\phi^3) + \frac{g^2}{2} \left(\int \text{tr}(\phi^3) \right)^2 + \dots \right)$$

$$\langle \varphi_{i_1}^{i_1}(P_1) \varphi_{i_2}^{i_2}(P_2) \varphi_{i_3}^{i_3}(P_3) \rangle$$

11

exhibit of $e^{\int \text{tr} \phi^3}$

$$\approx \frac{\int D\varphi e^{-S_0[\varphi]} (1 + \int \text{tr}(\phi^3)) (\varphi_{i_1}^{i_1}(P_1) \varphi_{i_2}^{i_2}(P_2) \varphi_{i_3}^{i_3}(P_3))}{\int D\varphi e^{-S_0[\varphi]}}$$

$$\phi_{i_4}^{i_4} \phi_{i_5}^{i_5} \phi_{i_6}^{i_6}$$

$$\langle \varphi_{i_1}^{i_1}(P_1) \varphi_{i_2}^{i_2}(P_2) \varphi_{i_3}^{i_3}(P_3) \rangle \approx$$

$$\phi_{i_4}^{i_4} \phi_{i_5}^{i_5} \phi_{i_6}^{i_6} \delta_{i_4}^{i_6} \delta_{i_5}^{i_4} \delta_{i_6}^{i_5}$$

$$\langle \varphi_{i_1}^{i_1}(P_1) \varphi_{i_2}^{i_2}(P_2) \varphi_{i_3}^{i_3}(P_3) \int dP_4 dP_5 dP_6 \text{tr}(\phi(P_4) \phi(P_5) \phi(P_6)) \delta(P_4 + P_5 + P_6) \rangle_0$$

$$\varphi_{i_1}^{i_1}(P_1)$$

$$\langle 1 2 3 4 5 6 \rangle =$$

$$\begin{aligned} &\langle 12 \rangle \langle 34 \rangle \langle 56 \rangle \\ &\langle 12 \rangle \langle 35 \rangle \langle 46 \rangle \\ &\langle 13 \rangle \quad \quad \quad \end{aligned}$$

$$\rightarrow \langle 12 \rangle \langle 34 \rangle \langle 56 \rangle$$

$$(6-1)!!$$

$$1 \quad 2 \quad 3$$

$$1 \quad 2 \quad 3 \rightarrow \text{connect } 456 \text{ among themselves}$$

$$1 \quad 2 \quad 3$$

$$1 \quad 2 \quad 3 \quad 4 \quad 5 \quad 6$$

$$\langle 14 \rangle \langle 25 \rangle \langle 36 \rangle \delta_{i_4}^{i_6} \delta_{i_5}^{i_4} \delta_{i_6}^{i_5}$$

$$\langle 14 \rangle \langle 26 \rangle \langle 35 \rangle$$

$$\langle 15 \rangle \langle 24 \rangle \langle 36 \rangle$$

$$\langle 15 \rangle \langle 26 \rangle \langle 34 \rangle$$

$$\langle 16 \rangle \langle 24 \rangle \langle 35 \rangle$$

$$\langle 16 \rangle \langle 25 \rangle \langle 34 \rangle$$

double empty

$$\text{tr}(AB) = \sum_{i,j} A^i_j B^j_i$$

$$(AB)^i_j = \sum_k A^i_k B^k_j$$

$$\text{tr}(ABC) = \sum_{i,j,k} A^i_j B^j_k C^k_i$$

$$= \sum_{\substack{i_1 j_1 \\ i_2 j_2 \\ i_3 j_3}} A^{i_1}_{j_1} B^{j_2}_{i_2} C^{i_3}_{j_3} \begin{pmatrix} i_1 & j_2 & i_3 \\ i_2 & i_3 & j_1 \end{pmatrix}$$

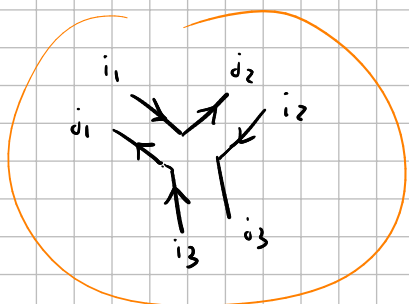
A1B1C

$$\sum_{\substack{i_4 j_4 \\ i_5 j_5 \\ i_6 j_6}} \int dP_4 dP_5 dP_6 \underbrace{\langle 14 \rangle \langle 25 \rangle \langle 36 \rangle}_{\delta^{i_1}_{j_4} \delta^{j_2}_{i_5} \delta^{i_3}_{j_6}} \delta(P_4 + P_5 + P_6) \delta^{j_6}_{i_4} \delta^{i_4}_{j_5} \delta^{j_5}_{i_6} \leftarrow \text{bar}$$

$$\sum_{i_4} \delta^{i_1}_{i_4} \delta^{j_6}_{i_4} = \delta^{i_1}_{j_6}$$

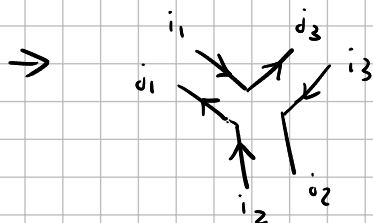
$$\sum_{\substack{i_4 j_4 \\ i_5 j_5 \\ i_6 j_6}} \delta^{i_1}_{j_4} \delta^{j_4}_{i_5} \delta^{i_2}_{j_5} \delta^{j_5}_{i_6} \delta^{i_3}_{j_6} \delta^{j_6}_{i_1} = \delta^{i_1}_{j_2} \delta^{j_2}_{i_3} \delta^{i_3}_{j_1}$$

$$\sum_{\substack{i_4 j_4 \\ i_5 j_5 \\ i_6 j_6}} \delta^{i_1}_{j_4} \delta^{j_4}_{i_5} \delta^{i_2}_{j_5} \delta^{j_5}_{i_6} \delta^{i_3}_{j_6} \delta^{j_6}_{i_1} = \sum_{\substack{i_4 j_4 \\ i_5 j_5 \\ i_6 j_6}} \delta^{i_3}_{j_1} \delta^{j_1}_{i_4} \delta^{i_1}_{j_4} \delta^{j_4}_{i_5} \delta^{i_2}_{j_5} \delta^{j_5}_{i_6} \delta^{i_3}_{j_6}$$



$\langle 14 \rangle \langle 26 \rangle \langle 35 \rangle$

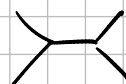
non cyclic permutation of previous one
swapping 2 and 3



$$A_3 = \sum_{S_n/Z_n} \text{of amplitude with fixed ordering}$$

$$n=4$$

$$\langle \underbrace{1\ 2\ 3\ 4}_{\text{external legs}} \underbrace{5\ 6\ 7}_{\text{tr}(\phi^3)} \underbrace{8\ 9\ 10}_{\text{tr}(\phi^3)} \rangle$$



cyclic symmetry $\overleftrightarrow{567}$ $\overleftrightarrow{8910}$

this tree if we vert contracted, the level things, we need to connect one of 567 with one of 8910

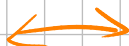
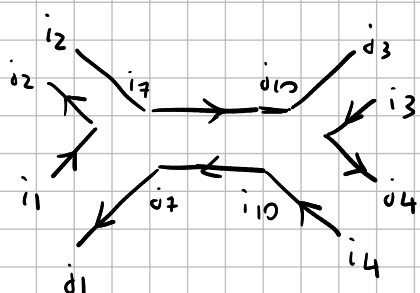
fix connection between 7 and 10

the rest needs to be created without loops

$$\langle 1\ 5 \rangle \langle 2\ 6 \rangle \langle 3\ 8 \rangle \langle 4\ 9 \rangle \langle 7\ 10 \rangle$$

• permutation of 1 2 3 4

$$\begin{array}{c} \delta_{i_4 i_9} \delta_{i_5 i_6} \\ \uparrow \\ \langle 1\ 5 \rangle \langle 2\ 6 \rangle \langle 3\ 8 \rangle \langle 4\ 9 \rangle \langle 7\ 10 \rangle \delta_{i_7 i_5} \delta_{i_6 i_7} \delta_{i_8 i_9} \delta_{i_9 i_{10}} \delta_{i_{10} i_8} \delta_{i_{10} i_7} \\ \downarrow \quad \downarrow \quad \downarrow \\ \delta_{i_1 i_5} \delta_{i_5 i_1} \quad \delta_{i_2 i_6} \delta_{i_6 i_5} \quad \delta_{i_7 i_{10}} \delta_{i_{10} i_7} \end{array}$$



fixed color ordering

w/o colors

$$\begin{array}{c} 2 \\ \diagup \quad \diagdown \\ 1 \quad 4 \end{array} \equiv \begin{array}{c} 2 \\ \diagup \quad \diagdown \\ 1 \quad 3 \end{array}$$

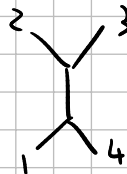
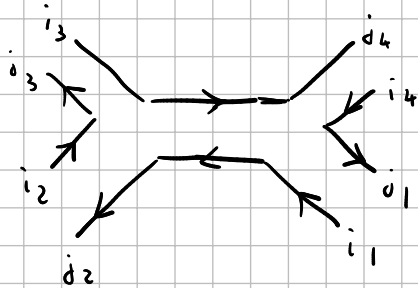
$$\langle 15 \rangle \langle 26 \rangle \langle 38 \rangle \langle 49 \rangle \langle 710 \rangle$$

↓

cyclic permutation

other signs

$$\langle 25 \rangle \langle 36 \rangle \langle 48 \rangle \langle 19 \rangle \langle 710 \rangle$$

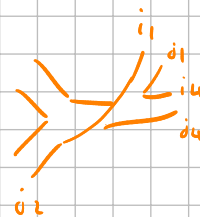


$$1 \rightarrow 2$$

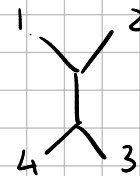
$$2 \rightarrow 3$$

$$3 \rightarrow 4$$

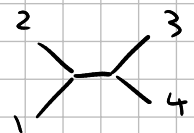
$$4 \rightarrow 1$$



$$\langle 35 \rangle \langle 46 \rangle \langle 18 \rangle \langle 29 \rangle \langle 710 \rangle$$



=



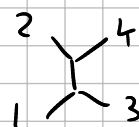
in the paper $T_1(T^{\sigma_{12}} T^{\sigma_{34}})$

$$A_4 = \sum_{\sigma \in S_4/\mathbb{Z}_4} \delta_{i_1 d_1}^{i_2} \delta_{i_2 d_2}^{i_3} \delta_{i_3 d_3}^{i_4} \delta_{i_4 d_4}^{i_1} A_4(\sigma(1) \dots \sigma(4))$$

nonzero amplitudes (planar)
(partial amplitude)

$$A_4(1234) = \text{diagram 1} + \text{diagram 2}$$

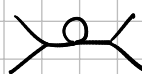
→ this includes

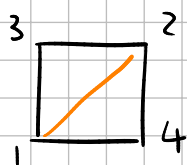
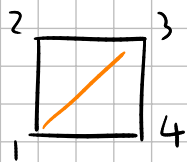


(u-channel)



factor of N





↙ partial ordered amplitudes

Res $A_n(123 \dots n)$ w.r.t one of the X_{ij} (the poles)

factorizes as product of two smaller amplitudes

if $f(z)$ is a complex function with Laurent series

$$\sum_{n \in \mathbb{Z}} a_n z^n$$

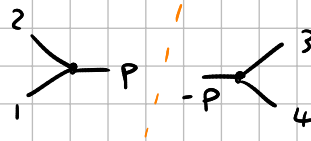
$$\text{Res } f = a_{-1}$$



$$A_4(1234) \propto \frac{1}{s_{12}} + \frac{1}{s_{23}}$$

look at residue near one of the poles (ex: s_{12})

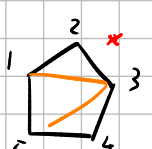
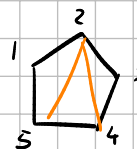
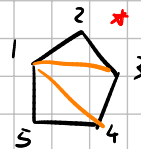
$$\text{Res}(A_4)_{s_{12}} = 1$$



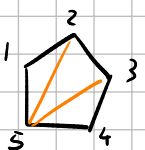
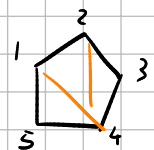
$$\propto \int dp A_3(12p) A_3(-p34)$$

$$A_4 = \frac{1}{s_{12}} \boxed{A_{\text{left}} A_{\text{right}}} + o(1)$$

if we want to expand at $s_{12} = X_{13}$

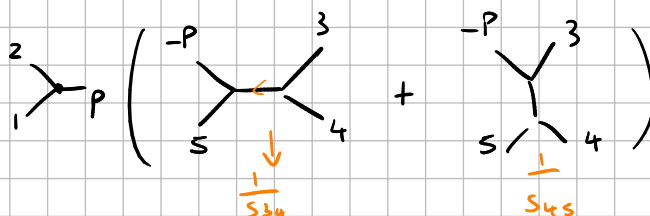


$$A_5(12345) = \boxed{\frac{1}{x_{13}x_{14}} + \frac{1}{x_{13}x_{35}}} + \frac{1}{x_{24}x_{25}} + \frac{1}{x_{24}x_{14}} + \frac{1}{x_{25}x_{35}}$$



$$\text{Res}(A_5)_{x_{13}} = \frac{1}{x_{14}} + \frac{1}{x_{35}} =$$

$$\begin{aligned} & \text{orange arrows} \\ & (p_1 + p_2 + p_3)^2 \\ & = (p_4 + p_5)^2 \end{aligned}$$



$$= A_3(12p) A_4(-p345)$$

this follows from $S^+ S = 11$ (S -matrix)

then amplitude A_n

choose 2 momenta i, j

$$P_i \rightarrow P_i + zq$$

$$z \in \mathbb{C}$$

$$P_j \rightarrow P_j - zq$$

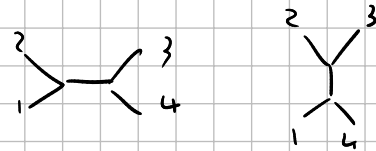
q is a null vector s.t. $P_i \cdot q = P_j \cdot q = q^2 = 0$

this does not affect that $P_i^2 = P_j^2 = 0$ and does not change S_{ij}

$$(P_i + zq)^2 = P_i^2 + \underbrace{2zq \cdot P_i}_{=0} + \underbrace{z^2 q^2}_{=0} = P_i^2$$

the new amplitude depends on z

$$A(n) = \oint dz \frac{A(z)}{z}$$



ex: $A_4(1234) = \frac{1}{S_{12}} + \frac{1}{S_{23}}$

$$P_1 \rightarrow P_1 + qz$$

$$P_2 \rightarrow P_2 - qz$$

$$S_{12} \rightarrow S_{12} = (P_1 + P_2)^2$$

$$S_{23} \rightarrow (P_2 - qz + P_3)^2$$

$$A_4(1234)(z) = \frac{1}{(P_1 + P_2)^2} + \frac{1}{(P_2 - qz + P_3)^2}$$

$A_4(n)$ = original amplitude

$$\frac{A(z)}{z} = \frac{A(0)}{z} + A'(0) + \frac{A''(0)}{2} z + \dots$$

$\frac{A(z)}{z}$ pole at $z=0$

expand in a series: $\text{Res} \frac{A(z)}{z} = A(0)$

$$A(n) = \text{Res} \frac{A(z)}{z} \propto \oint \frac{A(z)}{z} dz$$

$$s_{14} = (p_1 + p_4)^2 =$$

$$\text{look at } A_4(1234) = \frac{1}{s_{12}} + \frac{1}{s_{14}} = \frac{1}{s_{12}} - \frac{1}{s_{12} + s_{13}}$$

$$s_{12} + s_{13} = (p_1 + p_2)^2 + (p_1 + p_3)^2 = 2p_1 \cdot p_2 + 2p_1 \cdot p_3 = 2p_1 \cdot (p_2 + p_3)$$

$$= 2p_1(p_1 + p_2 + p_3) = 2p_1 \cdot (-p_4) = -(p_1 + p_4)^2 = -s_{14}$$

$$\text{if } s_{13} = 0 \quad \text{then} \quad A_4(1234) = 0$$

$$A_4 = \frac{1}{2p_1 \cdot p_2} + \frac{1}{2p_1 \cdot p_4}$$

$$p_1 \cdot p_3 = 0$$

$$p_1 \cdot (p_1 + p_2 + p_3 + p_4) = 0$$

↓

$$p_1 \cdot p_2 = -p_1 \cdot p_4$$

$$A_5 = \frac{1}{x_{13} x_{14}} + \frac{1}{x_{13} x_{35}} + \frac{1}{x_{24} x_{25}} + \frac{1}{x_{24} x_{14}} + \frac{1}{x_{25} x_{35}}$$

$$= \frac{1}{\underline{s_{12}} \underline{s_{45}}} + \frac{1}{\underline{s_{23}} \underline{s_{15}}} + \frac{1}{\underline{s_{34}} \underline{s_{21}}} + \frac{1}{\underline{s_{45}} \underline{s_{32}}} + \frac{1}{\underline{s_{51}} \underline{s_{43}}}$$

$$p_1 + p_2 + p_3 + p_4 + p_5 = 0$$

$$p_i^2 = 0$$

$$s_{i0} = 2p_i \cdot p_0$$

$$\frac{1}{s_{12}} \left(\frac{1}{s_{45}} + \frac{1}{s_{34}} \right) + \frac{1}{s_{23}} \left(\frac{1}{s_{15}} + \frac{1}{s_{45}} \right) = -\frac{1}{s_{15} s_{43}}$$

$$\frac{1}{s_{12}} \left(\frac{s_{15} s_{43}}{s_{45}} + s_{15} \right) + \frac{1}{s_{23}} \left(s_{43} + \frac{s_{15} s_{43}}{s_{45}} \right) = -1$$

$$s_{43} = 0$$

$$\frac{s_{15}}{s_{12}} = -1$$

$$s_{13} = s_{14} = 0$$

$$\frac{1}{s_{12} s_{45}} + \frac{1}{s_{23} s_{15}} + \frac{1}{s_{34} s_{21}} + \frac{1}{s_{45} s_{32}} + \frac{1}{s_{51} s_{43}}$$

$$p_1 \cdot p_3 = 0 \quad p_1 \cdot p_4 = 0$$

$$p_1 \cdot p_2 = -p_1 \cdot p_5$$

$$p_1 \cdot (p_2 + p_3 + p_4 \cdot p_5) = 0$$

$$s_{15} = -s_{12}$$

wxy	13	14
wxy	24	25
	31	35
	42	41
	52	53

$$\frac{1}{s_{12} s_{45}} - \frac{1}{s_{23} s_{12}} + \frac{1}{s_{34} s_{12}} + \frac{1}{s_{45} s_{32}} - \frac{1}{s_{12} s_{43}}$$

$$A_5 = \text{graph} + \text{cyclic}$$

$$= \frac{1}{s_{12} s_{34}} + \frac{1}{s_{23} s_{45}} + \frac{1}{s_{34} s_{51}} + \frac{1}{s_{45} s_{12}} + \frac{1}{s_{51} s_{23}}$$

$$p_1 \cdot p_3 = 0 \quad p_1 \cdot p_4 = 0$$

$$p_1 \cdot p_2 = -p_1 \cdot p_5$$

$$p_1 \cdot (p_2 + p_3 + p_4 \cdot p_5) = 0$$

$$s_{15} = -s_{12}$$

$$p_1 \cdot p_2 + p_1 \cdot p_5 = 0$$

$$\frac{1}{\cancel{S_{12}} \cancel{S_{34}}} + \frac{1}{S_{23} S_{45}} - \frac{1}{\cancel{S_{34}} S_{12}} + \frac{1}{S_{45} S_{12}} - \frac{1}{S_{12} S_{23}}$$

$$\frac{1}{S_{23} S_{45}} + \frac{1}{S_{45} S_{12}} - \frac{1}{S_{12} S_{23}}$$

$$1 + \frac{S_{23}}{S_{12}} - \frac{S_{45}}{S_{12}} \Rightarrow \frac{S_{12} + S_{23} - S_{45}}{S_{23} S_{12} S_{45}} = 0$$

$$P_2 \cdot (P_1 + P_3 + P_4 \cdot P_5) = 0$$

$$S_{12} + \cancel{S_{23}} + \cancel{S_{14}} + \cancel{S_{25}} = 0$$

$$P_i \left(\sum_j P_j \right) = 0 \Rightarrow \sum_j S_{ij} = 0$$

$$i=2 \quad \underline{S_{21}} + \underline{S_{23}} + \cancel{S_{24}} + \cancel{S_{25}} = 0 \quad (+)$$

$$i=4 \quad \cancel{S_{42}} + \cancel{S_{43}} + \underline{S_{45}} = 0 \quad (=)$$

$$i=3 \quad \underline{S_{32}} + \cancel{S_{34}} + \cancel{S_{35}} = 0 \quad (+)$$

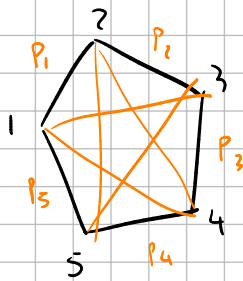
$$i=5 \quad \underline{\cancel{S_{12}}} + \cancel{S_{52}} + \cancel{S_{53}} + \underline{\cancel{S_{54}}} = 0 \quad (-)$$

$$\underline{S_{12}} + \underline{S_{23}} - \underline{S_{45}}$$

$$2(S_{21} + S_{23} - S_{45}) = 0$$

$$S_{13} = S_{14} = 0$$

$$S_{24} = S_{25} = 0$$



$$X_{13}$$

$$X_{14}$$

$$X_{24}$$

$$X_{25}$$

$$X_{35}$$

$$C_{i,j} = -S_{i,j} = -(P_i + P_j)^2$$

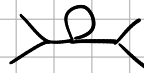
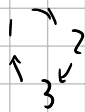
$$X_{13} = (P_1 + P_2)^2$$

→ planar

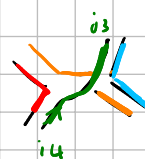
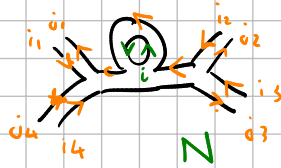
$$X_{25} = (P_2 + P_3 + P_4)^2$$

$$C_{12} = X_{13} \text{ planar}$$

$$C_{13} = (P_1 + P_3)^2 \text{ not planar if } n > 3$$



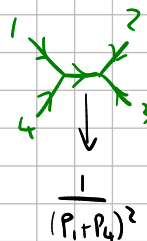
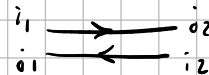
$$\langle \phi_{i1}^{i1} (p_1) \phi_{i2}^{i2} (p_2) \dots \rangle$$

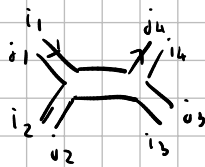
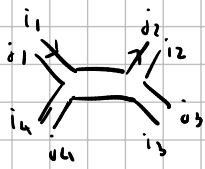


$$\mathcal{L} = \frac{1}{2} \text{tr}(\partial_\mu \phi \partial^\mu \phi) + V(\phi)$$

→ gives projector

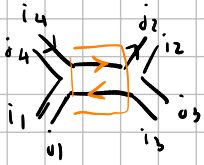
$$\sum_{\substack{i_1 i_2 \\ i_3 i_4}} \partial_\mu \phi_{i_1}^{i1} \partial^\mu \phi_{i_2}^{i2} \delta_{i_3}^{i1} \delta_{i_4}^{i2}$$



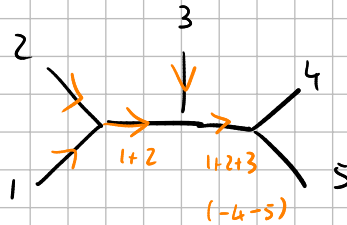


$$\frac{1}{(P_1 + P_4)^2} \delta_{i_2}^{j_2} \delta_{i_3}^{j_3} \dots$$

$$\frac{1}{(P_1 + P_6)^2} \delta_{i_4}^{j_4}$$

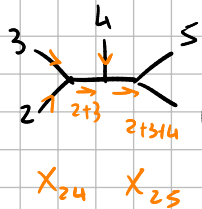
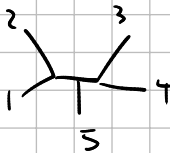


$$\frac{1}{(P_1 + P_4)^2} \delta_{i_4}^{j_4}$$



$$\frac{1}{X_{13}} \frac{1}{X_{14}} \frac{1}{S_{12}} \frac{1}{S_{123}}$$

$$A_5(1, 2, 3, 4, 5) = \frac{1}{X_{13} X_{14}} + \text{cyclic}$$



$$\frac{1}{X_{24} X_{25}} + \frac{1}{X_{35} X_{31}} + \dots$$

$$\frac{1}{S_{12}} \frac{1}{S_{12}}$$

$$X_{24} X_{25}$$



$P \cdot P$

$$P^2 = 0$$

$$\Rightarrow E^2 = \vec{P}^2 \leftarrow \text{energy of a photon}$$

Minkowski space:

$$P_\mu P^\mu = 0$$

$$P_t P_t - P_x P_x - P_y P_y - P_z P_z = 0$$

$$P_t = E \text{ energy}$$

$$\vec{P} = (P_x, P_y, P_z) \text{ momentum}$$

$$A_4 \propto \frac{1}{S_{12}} + \frac{1}{S_{13}} + \frac{1}{S_{14}} =$$

$$\frac{1}{S_{13}} \text{ Acft Aright}$$

$$f(z) = \frac{a_{-1}}{z} + a_0 + a_1 z + a_2 z^2 + \dots$$

$$\text{Res } f(z) = a_{-1}$$

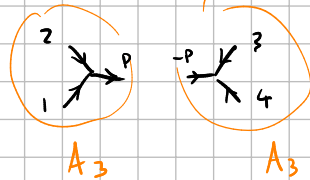
$$A_4 = \text{diagram 1} + \text{diagram 2}$$

$$= \frac{1}{s_{14}} + \frac{1}{s_{12}} \rightarrow O(1) \text{ w.r.t. } s_{12}$$

$$S^+ = S$$

$\Rightarrow$ residue factorization

$$\text{Res}_{s_{12}=0} A_4 \propto 1$$



4-momentum $p^0 = E(\vec{p})$

non-local: the QFT cannot be built with path integrals of an action
(canonical quantization starting from Lagrangian)

$$\text{diagram 1} = \text{diagram 2} + \text{diagram 3} + \text{diagram 4} + \text{diagram 5} + \dots$$

$$f(z, w) = \frac{1}{z} + \frac{1}{w} + \frac{z}{w^2}$$

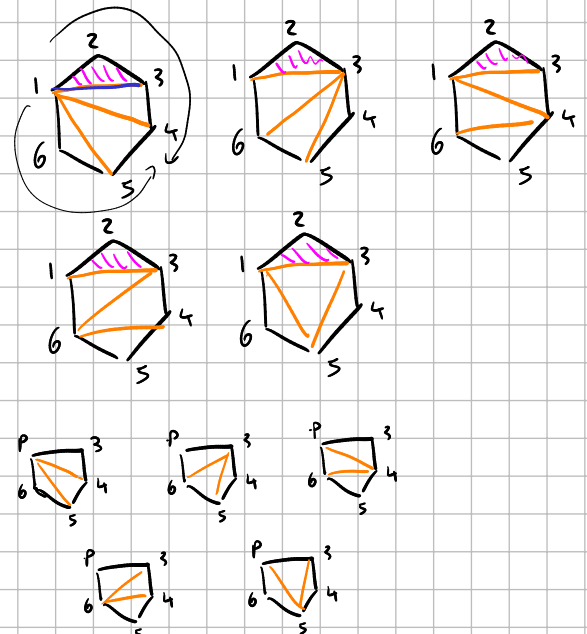
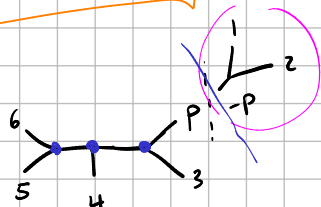
expand in powers of $\frac{1}{z}$

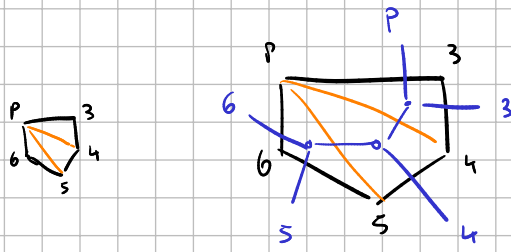
$$f(z, w) = \frac{1}{z} + O(1) \Leftrightarrow \text{Res } f_{z=0} = 1$$

$\text{Res}(A_6)$ look at residue at x_{13}

$$A_6 = \frac{1}{x_{13}} \left(\frac{1}{x_{14} x_{15}} + \frac{1}{x_{36} x_{35}} + \frac{1}{x_{14} x_{46}} + \frac{1}{x_{36} x_{46}} + \frac{1}{x_{35} x_{15}} \right) + O(1)$$

$$\frac{1}{(p_4 + p_5 + p_6)^2 (p_5 + p_6)^2}$$



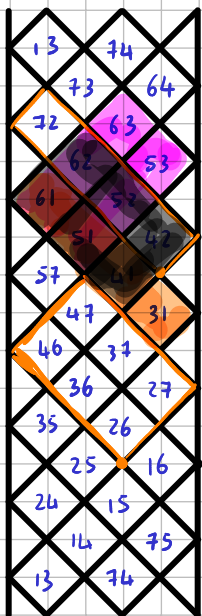
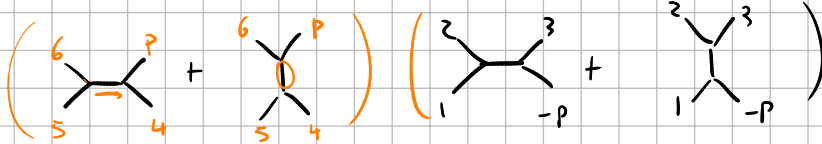
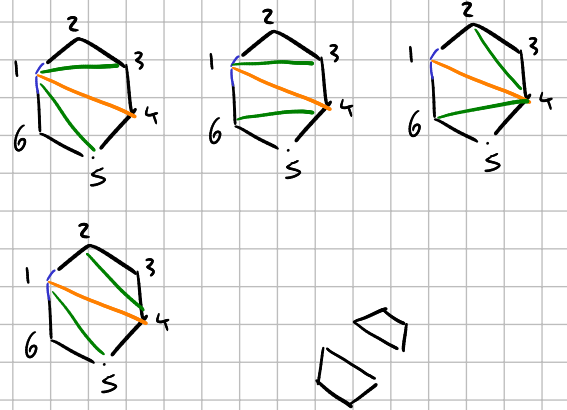


$$A_6 = \frac{1}{x_{14}} \left(\frac{1}{x_{13} x_{15}} + \frac{1}{x_{13} x_{40}} + \frac{1}{x_{24} x_{40}} + \frac{1}{x_{24} x_{15}} \right)$$

$$= \frac{1}{x_{14}} \left(\frac{1}{x_{15}} \left(\frac{1}{x_{13}} + \frac{1}{x_{24}} \right) + \frac{1}{x_{40}} \left(\frac{1}{x_{13}} + \frac{1}{x_{24}} \right) \right)$$

$$= \frac{1}{x_{14}} \left(\left(\frac{1}{x_{15}} + \frac{1}{x_{40}} \right) \left(\frac{1}{x_{13}} + \frac{1}{x_{24}} \right) \right)$$

$s_{56} + s_{45}$



→ 1 zero

$$C_{42} = C_{52} = C_{62} = C_{72} = 0$$

→ 2-zero

$$C_{26} = C_{36} = C_{46} = 0$$

$$C_{27} = C_{37} = C_{47} = 0$$

→

$$C_{6j} = 0$$

$$C_{7j} = 0$$

$j = 3, \dots, 13$

$$C_{14} = C_{15} = C_{16} = 0$$

$$C_{24} = C_{25} = C_{26} = 0$$

$$1234567 \rightarrow 6712345$$

$$C_{62} = C_{63} = C_{64} = 0$$

$$C_{12} = C_{13} = C_{14} = 0$$

1-zero:

$$C_{1j} = 0 \quad j = 3, \dots, n-1$$

2-zero

$$\begin{cases} C_{1j} = 0 \\ C_{2j} = 0 \end{cases} \quad j = 4, \dots, n-1$$

3-zero

$$\begin{cases} C_{1j} = 0 \\ C_{2j} = 0 \\ C_{3j} = 0 \end{cases} \quad j = 5, \dots, n-1$$

k-zero:

$$\begin{cases} C_{1j} = 0 \\ C_{2j} = 0 \\ \vdots \\ C_{kj} = 0 \end{cases} \quad j = k+2, \dots, n-1$$

$$\sum_j c_{ij} = 0 \quad \text{with 1-zero} \Rightarrow S_{i2} = -S_{in}$$

$$c_{ij} = X_{ij} + X_{in, j+1} - X_{i+1, j} - X_{i, j+1}$$

$$n=7$$

$$k=3:$$

$$j=5, 6$$

$$c_{15} = c_{16} = 0$$

$$c_{25} = c_{26} = 0$$

$$\vdots$$

$$c_{75} = c_{76} = 0$$

$$1 \leq i < j \leq n$$

$$X_{ij} = \left(\sum_{k=i}^{j-1} p_k \right)^2$$

Q: which X_{ij} are affected by setting $c_{ij}=0$ for $j=3, \dots, n-1$?

$$c_{ij} = -(p_i + p_j)^2$$

$$c_{ij} = -(p_i + p_j)^2 = -2p_i \cdot p_j$$

$X_{1, j+1}$ depends on p_1

"

$$\left(\sum_{k=1}^j p_k \right)^2 = \underbrace{(p_1 + p_2 + \dots + p_j)}_{A+B}^2 \quad A^2 + 2AB + B^2$$

$$= \underbrace{(p_1 + p_2)^2}_{S_{12}} + \underbrace{2(p_1 + p_2) \cdot (p_3 + \dots + p_j)}_{S_{3,4,\dots,j}} + (p_3 + \dots + p_j)^2$$

$$2(p_1 + p_2) \cdot (p_3 + \dots + p_j) = 2p_1 \cdot (p_3 + \dots + p_j) + 2p_2 \cdot (p_3 + \dots + p_j)$$

$$= \sum_{i=3}^j \underbrace{2p_1 \cdot p_i}_{S_{1i}} + \sum_{i=3}^j \underbrace{2p_2 \cdot p_i}_{S_{2i}}$$

$$S_{i_1 i_2 \dots i_n} = (p_{i_1} + p_{i_2} + \dots + p_{i_n})^2$$

$$\int d^3p |p\rangle\langle p| = \mathbb{1}$$

$$S^\dagger = S = \lim_{T \rightarrow \infty} U(T, -T)$$

$$S = \mathbb{1} + iT$$

$$S^\dagger S = \mathbb{1} \rightarrow \mathbb{1} + iT^\dagger - iT + T^\dagger T = \mathbb{1}$$

$$\langle 0 | T | p_1 p_2 \dots p_n \rangle = A_n$$

$$A_4: \begin{array}{c} 1 \quad 2 \\ \diagdown \quad \diagup \\ \text{---} \\ \diagup \quad \diagdown \\ 4 \quad 3 \end{array} + \begin{array}{c} 1 \quad 2 \\ \diagdown \quad \diagup \\ \text{---} \\ \diagup \quad \diagdown \\ 4 \quad 3 \end{array}$$

$$\text{Res } A_4(s_{12}=0) = s_{12} \begin{array}{c} 1 \quad 2 \\ \diagdown \quad \diagup \\ \text{---} \\ \diagup \quad \diagdown \\ 4 \quad 3 \end{array} = 1$$

$$\text{Res } A_5(s_{12}=0) = \begin{array}{c} 1 \quad 2 \\ \diagdown \quad \diagup \\ \text{---} \\ \diagup \quad \diagdown \\ 4 \quad 3 \end{array} + \begin{array}{c} 1 \quad 2 \\ \diagdown \quad \diagup \\ \text{---} \\ \diagup \quad \diagdown \\ 4 \quad 3 \end{array} = \underbrace{\begin{array}{c} 1 \quad 2 \\ \diagdown \quad \diagup \\ \text{---} \\ \diagup \quad \diagdown \\ P \quad 3 \end{array}}_{A_3} \left(\underbrace{\begin{array}{c} 6 \quad 2 \\ \diagdown \quad \diagup \\ \text{---} \\ \diagup \quad \diagdown \\ 5 \quad 4 \quad 3 \end{array}}_{A_4} + \begin{array}{c} 6 \quad 2 \\ \diagdown \quad \diagup \\ \text{---} \\ \diagup \quad \diagdown \\ 5 \quad 4 \quad 3 \end{array} \right)$$

$$\text{Res } A_6(s_{123}=0) = \begin{array}{c} 6 \quad 2 \\ \diagdown \quad \diagup \\ \text{---} \\ \diagup \quad \diagdown \\ 5 \quad 4 \quad 3 \end{array} \begin{array}{c} 1 \quad 2 \\ \diagdown \quad \diagup \\ \text{---} \\ \diagup \quad \diagdown \\ P \quad 3 \end{array} = \begin{array}{c} 6 \quad 2 \\ \diagdown \quad \diagup \\ \text{---} \\ \diagup \quad \diagdown \\ 5 \quad 4 \quad 3 \end{array} \begin{array}{c} 1 \quad 2 \\ \diagdown \quad \diagup \\ \text{---} \\ \diagup \quad \diagdown \\ P \quad 3 \end{array} + \begin{array}{c} 6 \quad 2 \\ \diagdown \quad \diagup \\ \text{---} \\ \diagup \quad \diagdown \\ 5 \quad 4 \quad 3 \end{array} \begin{array}{c} 1 \quad 2 \\ \diagdown \quad \diagup \\ \text{---} \\ \diagup \quad \diagdown \\ P \quad 3 \end{array} = \begin{array}{c} 1 \quad 2 \\ \diagdown \quad \diagup \\ \text{---} \\ \diagup \quad \diagdown \\ P \quad 3 \end{array} \left(\begin{array}{c} 6 \quad 2 \\ \diagdown \quad \diagup \\ \text{---} \\ \diagup \quad \diagdown \\ 5 \quad 4 \quad 3 \end{array} + \begin{array}{c} 6 \quad 2 \\ \diagdown \quad \diagup \\ \text{---} \\ \diagup \quad \diagdown \\ 5 \quad 4 \quad 3 \end{array} \right) + \begin{array}{c} 1 \quad 2 \\ \diagdown \quad \diagup \\ \text{---} \\ \diagup \quad \diagdown \\ P \quad 3 \end{array} \left(\begin{array}{c} 6 \quad 2 \\ \diagdown \quad \diagup \\ \text{---} \\ \diagup \quad \diagdown \\ 5 \quad 4 \quad 3 \end{array} + \begin{array}{c} 6 \quad 2 \\ \diagdown \quad \diagup \\ \text{---} \\ \diagup \quad \diagdown \\ 5 \quad 4 \quad 3 \end{array} \right)$$

$$= \underbrace{\begin{array}{c} 6 \quad 2 \\ \diagdown \quad \diagup \\ \text{---} \\ \diagup \quad \diagdown \\ 5 \quad 4 \quad 3 \end{array}}_{A_4(456-P)} \begin{array}{c} 1 \quad 2 \\ \diagdown \quad \diagup \\ \text{---} \\ \diagup \quad \diagdown \\ P \quad 3 \end{array} + \begin{array}{c} 6 \quad 2 \\ \diagdown \quad \diagup \\ \text{---} \\ \diagup \quad \diagdown \\ 5 \quad 4 \quad 3 \end{array} \begin{array}{c} 1 \quad 2 \\ \diagdown \quad \diagup \\ \text{---} \\ \diagup \quad \diagdown \\ P \quad 3 \end{array} \left(\begin{array}{c} 1 \quad 2 \\ \diagdown \quad \diagup \\ \text{---} \\ \diagup \quad \diagdown \\ P \quad 3 \end{array} + \begin{array}{c} 1 \quad 2 \\ \diagdown \quad \diagup \\ \text{---} \\ \diagup \quad \diagdown \\ P \quad 3 \end{array} \right)$$

BCFW shift: pick p_i, p_j among the external momenta ($i \neq j$)

send $p_i \rightarrow p_i + zq$ momentum q s.t. $q^2 = 0$
 $p_j \rightarrow p_j - zq$ $q \cdot p_i = q \cdot p_j = 0$

ex: 4 particles momenta: p_1, p_2, p_3, p_4

after shift: $p_1 + zq, p_2 - zq, p_3, p_4$

• total momentum is still zero

• $(p_1 + zq)^2 = (p_2 - zq)^2 = 0$

amplitudes now depend on z

$$A_4 = \begin{array}{c} 1 \\ \diagup \quad \diagdown \\ 4 \quad 3 \end{array} + \begin{array}{c} 1 \\ \diagdown \quad \diagup \\ 4 \quad 3 \end{array} = \frac{1}{s_{12}} + \frac{1}{s_{14}} = \frac{1}{p_1 \cdot p_2} + \frac{1}{p_1 \cdot p_4}$$

after shift: $A_4(z) = \frac{1}{(p_1 + zq)(p_2 - zq)} + \frac{1}{(p_1 + zq) \cdot p_4} = \frac{1}{p_1 \cdot p_2} + \frac{1}{p_1 \cdot p_4 + z p_4 \cdot q}$
 $\Rightarrow = p_1 \cdot p_2$

look at $\frac{A(z)}{z}$ pole at $z=0$

$$A(0) = 2\pi i \oint_{z=0} \frac{A(z)}{z} dz$$

small loop around $z=0$

$$\begin{aligned} X_{1, j+1} &= s_{12 \dots j} = (p_1 + p_2 + \dots + p_j)^2 = (p_1 + p_2)^2 + (p_3 + p_4 + \dots)^2 + 2(p_1 + p_2) \cdot (p_3 + p_4 + \dots) \\ &= s_{12} + s_{3 \dots j} + 2p_1 \cdot (p_3 + p_4 + \dots) + 2p_2 \cdot (p_3 + p_4 + \dots) \\ &= s_{12} + 2 \sum_{i=3}^j p_1 \cdot p_i = s_{1i} + 2 \sum_{i=3}^j p_2 \cdot p_i = s_{2i} + s_{3 \dots j} \\ &= s_{12} + \sum_{i=3}^j s_{1i} + \sum_{i=3}^j s_{2i} + s_{3 \dots j} \end{aligned}$$

$$X_{2,i+1} = (p_2 + \dots + p_i)^2 = (p_3 + p_4 + \dots)^2 + 2 p_2 \cdot (p_3 + p_4 + \dots)$$

$$= \sum_{i=3}^n s_{2,i} + s_{3,\dots,i}$$

$$X_{2,n} = s_{2,3,\dots,n-1} = (p_2 + p_3 + \dots + p_{n-1})^2 = (p_1 + p_n)^2 = 2 p_1 \cdot p_n$$

$$= 2 p_1 \cdot (-p_1 - p_2 - \dots - p_{n-1})$$

$$= -2 p_1 \cdot p_2 - 2 p_1 (p_3 + p_4 + \dots + p_{n-1})$$

$$= -s_{1,2} - \sum_{i=3}^{n-1} s_{1,i}$$

(A) set $s_{1,i} = 0$ ($i=3, \dots, n-1$)

(B) $p_2 \rightarrow p_2 + qz$
 $p_n \rightarrow p_n - qz$

(A₅) $X_{1,3} \quad X_{1,4} \quad X_{2,4} \quad X_{2,5} \quad X_{3,5} \quad \left(\begin{array}{l} X_{ii} = 0 \\ X_{ij} = X_{ji} \end{array} \right)$

shift: $p_2 \rightarrow p_2 + qz \quad p_5 \rightarrow p_5 - qz$

affected if $X_{i,j}$ contains either only p_2 or only p_5

$X_{1,3}$	affected	$(p_1 + p_2)^2$
$X_{1,4}$	affected	$(p_1 + p_2 + p_3)^2 = (p_4 + p_5)^2$
$X_{2,4}$	affected	$(p_2 + p_3)^2$
$X_{2,5}$	affected	$(p_2 + p_3 + p_4)^2$
$X_{3,5}$	unaffected	$(p_3 + p_4)^2 = (p_1 + p_2 + p_5)^2$

note generally: if p_2 and p_n shifted:

$X_{1,i+1}$ always only contains p_2

$X_{2,i+1}$ always only contains p_2

$X_{3,i+1}$ contains no p_2 and no p_n

$$A_5 \quad X_{13} \quad X_{14} \quad X_{24} \quad X_{25} \quad X_{35}$$

$$\text{zero: } S_{13} = S_{14} = 0$$

$$X_{13} = (p_1 + p_2)^2 = S_{12} \quad \text{not effected}$$

$$X_{14} = (p_1 + p_2 + p_3)^2 = \boxed{S_{123}} \xrightarrow{\text{effected}} = \boxed{S_{45}}$$

$$= (\cancel{p_1 + p_3})^2 + 2p_2 \cdot (p_1 + p_3) = \boxed{S_{12} + S_{23}}$$

$\rightarrow S_{13} = 0$

$$X_{24} = (p_2 + p_3)^2 = S_{23} \quad \text{not effected}$$

$$= (\underbrace{p_4 + p_5 + p_1}_{=0})^2 = (p_1 + p_4)^2 + 2p_5(p_1 + p_4) = S_{15} + S_{45}$$

$$X_{35} = (p_3 + p_4)^2 = S_{34}$$

$$= (p_1 + p_2 + p_5)$$

$$S_{13} = S_{14} = S_{15} = 0$$

$$n=6$$

$$X_{36} = (p_3 + p_4 + p_5)^2 = (p_1 + p_2 + p_6)$$

$$X_{35} = (p_3 + p_4)^2 = (p_1 + p_2 + p_5 + p_6)$$

$$X_{i,j+1}^{(0)} \text{ is the } X_{i,j+1} \text{ after setting } S_{1i} = 0 \quad (i=3, \dots, n-1)$$

$$2 \leq j \leq n-2$$

$$X_{1,j+1}^{(0)} = S_{12} + \sum_{i=3}^j S_{2i} + S_{3 \dots j}$$

$$X_{2,j+1}^{(0)} = \sum_{i=3}^j S_{2i} + S_{3 \dots j}$$

$$X_{2n}^{(0)} = -S_{12}$$

$$\text{after BCFW shift} \quad X_{ij} = X_{ij}(z) \quad (\text{polynomial})$$

$$X_{ij}^{(\infty)} \text{ is the leading order coefficient in } z$$

$$X_{ij}^{(\infty)} = \left. \frac{d}{dz} \right|_{z=0} X_{ij}(z) \quad || \quad \text{or} \quad X_{ij}(z) = X_{ij}^{(0)} + \boxed{X_{ij}^{(\infty)}} z + O(z^2)$$

shift p_2 and p_n

$$X_{i+1}(z) = s_{12}(z) + \cancel{\sum_{i=3}^d s_{1i}(z)} + \cancel{\sum_{i=3}^d s_{2i}(z)} + s_{3 \dots j}(z) = s_{12}(z) + \sum_{i=3}^d s_{2i}(z) + o(1)$$

$$s_{12}(z) = z p_1 \cdot (p_2 + qz) = z z p_1 \cdot q + o(1)$$

$$s_{2i}(z) = z p_i \cdot (p_2 + qz) = z z p_i \cdot q + o(1)$$

$$X_{i+1}(z) = z z p_1 \cdot q + z z \sum_{i=3}^d p_i \cdot q + o(1)$$

$$\begin{cases} X_{i+1}^{(\infty)} = \underline{z q \cdot p_1} + \underline{z \sum_{i=3}^d q \cdot p_i} \\ X_{2i+1}^{(\infty)} = \underline{z \sum_{i=3}^d q \cdot p_i} \\ X_{2n}^{(\infty)} = \underline{-z q \cdot p_1} \end{cases}$$

$$i = 2, \dots, n-2$$

$$n-2-3+1 = n-3 \text{ eqs}$$

Relations:

$$X_{i+1}^{(\infty)} - X_{2i+1}^{(\infty)} + X_{2n}^{(\infty)} = 0$$

$$i = 2, \dots, n-2$$

$$X_{i+1}^{(0)} - X_{2i+1}^{(0)} + X_{2n}^{(0)} = 0$$

$$(n-3) \text{ eqs}$$

$$X_{i+1}(z) - X_{2i+1}(z) + X_{2n}(z) = o(1)$$

$$A_4 = a_1 \begin{array}{c} 1 \quad 2 \\ \diagdown \quad \diagup \\ \text{---} \\ \diagup \quad \diagdown \\ 4 \quad 3 \end{array} + a_2 \begin{array}{c} 1 \quad 2 \\ \diagdown \quad \diagup \\ \text{---} \\ \diagup \quad \diagdown \\ 4 \quad 3 \end{array}$$

$$s_{12} \quad s_{13} \quad s_{14} \quad s_{23} \quad s_{24} \quad s_{34}$$

impose the hidden zero condition $\Rightarrow a_1 = a_2$

$$\frac{1}{x}(\dots) + \frac{1}{y}(\dots)$$

goal: show that some of the X_{ij} variables satisfy the same relations if we set a 1-zero or perform a BCFW shift (up to order ϵ)

1-zero: $c_{13} = c_{14} = \dots = c_{1,n-1} = 0 \Rightarrow n-3$ constraints

started with $\frac{n(n-3)}{2}$ variables, now we have $n-3$ less of them

we know that $c_{ij} = X_{ij} + X_{i+1,j+1} - X_{i+1,j} - X_{i,j+1}$

$$0 = X_{13} + X_{24} - \overset{=0}{X_{23}} - X_{14}$$

$$\text{since } c_{13} = 0$$

$$0 = X_{14} + X_{25} - X_{24} - X_{15}$$

$\vdots$

$$0 = X_{1,n-1} + X_{2n} - X_{2,n-1} - \overset{=0}{X_{1n}}$$

$$X_{24} = X_{14} - X_{13}$$

$$X_{21} = X_{23} = 0$$

$$X_{25} = X_{24} - X_{14} + X_{15} = X_{15} - X_{13}$$

$$X_{26} = X_{16} - X_{13}$$

$\vdots$

$$X_{2n} = X_{1n} - X_{13} = -X_{13}$$

let's do shift instead

$$P_2 \rightarrow P_2 + qz$$

$$P_n \rightarrow P_n - qz$$

which X_{ij} are affected by the shift? (assume $i < j$)

$$(P_i + P_{i+1} + \dots + P_{j-1})^2$$

$$i=1 \text{ or } i=2$$

$$X_{1j} \quad j=3, \dots, n-1$$

$$X_{2j} \quad j=4, \dots, n$$

are the only affected ones

expand $X_{1j}(z)$ to linear order

$X_{2j}(z)$ to linear order

$$X_{1j}(z) = (P_1 + P_2 + qz + \dots + P_{j-1})^2$$

$$X_{2j}(z) = (P_2 + qz + \dots + P_{j-1})^2$$

$$X_{1j}(z) = (P_1 + P_2 + qz + \dots + P_{j-1})^2$$

$$= (qz + \underbrace{P_1 + P_2 + \dots + P_{j-1}}_{=0})^2$$

$$= (P_1 + P_2 + \dots + P_{j-1})^2 + \cancel{z^2 q^2} + 2zq \cdot (P_1 + P_2 + \dots + P_{j-1})$$

$$= X_{1j}(0) + \underbrace{2zq \cdot (P_1 + P_2 + \dots + P_{j-1})}_{\rightarrow \text{call it } X_{1j}^{(\infty)}} \rightarrow \text{call it } X_{1j}^{(\infty)}$$

$$= X_{1j}(0) + X_{1j}^{(\infty)} z$$

$$z \sim 1+z$$

$$X_{2j}(z) = (P_2 + qz + \dots + P_{j-1})^2$$

$$= X_{2j}(0) + \underbrace{2qz \cdot (P_3 + \dots + P_{j-1})}_{\rightarrow X_{2j}^{(\infty)}} \rightarrow X_{2j}^{(\infty)}$$

$$= X_{2j}(0) + X_{2j}^{(\infty)} z$$

$$X_{2j}^{(\infty)} = 2q \cdot (p_3 + \dots + p_{j-1})$$

$$X_{2j}^{(\infty)} = X_{1j}^{(\infty)} - X_{13}^{(\infty)}$$

$$X_{1j}^{(\infty)} = 2q \cdot (p_1 + p_3 + \dots + p_{j-1})$$

$$= \underbrace{2q \cdot p_1}_{\text{" } X_{13}^{(\infty)}} + X_{2j}^{(\infty)}$$

$$X_{2j}^{(\infty)} = X_{1j}^{(\infty)} - X_{13}^{(\infty)}$$

$$X_{24}^{(n)} = X_{14}^{(n)} - X_{13}^{(n)}$$

$$X_{25}^{(n)} = X_{15}^{(n)} - X_{13}^{(n)}$$

$$X_{26}^{(n)} = X_{16}^{(n)} - X_{13}^{(n)}$$

$$X_{2n}^{(n)} = X_{1n}^{(n)} - X_{13}^{(n)} = -X_{13}^{(n)}$$

let B be an homogeneous rational function of the X_{ij}

a function $f(\lambda x_1, \lambda x_2, \dots, \lambda x_n) = \lambda^n f(x_1, \dots, x_n)$

let B_i be the subset of all terms in B that under the shift
 $p_2 \rightarrow p_2 + 2q \quad p_n \rightarrow p_n - 2q$ scales as z^i as $z \rightarrow \infty$
 $i \in \mathbb{Z}$

ex:

$$B(z) = \frac{1}{X_{24}(z)}$$

$$= \frac{1}{X_{2j}(0) + X_{2j}^{(\infty)} z}$$

scales like $\frac{1}{z}$

$\Rightarrow B_{-1}$

$$B(z) = \underbrace{X_{25}(z) + X_{26}(z)}_{B_1(z)} + \underbrace{\frac{X_{35}^2}{X_{24}(z)}}_{B_{-1}(z)} + \underbrace{X_{35}}_{B_0(z)}$$

in general:
scaling

$$B = B_m + B_{m-1} + B_{m-2}$$

where z^m is the largest

Prop: these are equivalent:

①. B satisfies the 1-zero $\stackrel{\text{def}}{\iff} \left(\text{when } X_{2i} = X_{1i} - X_{13} \text{ holds} \right)$
 $B=0$

②. each B_i satisfies the 1-zero

③. each B_i has enhanced UV soly z^{i-1}

$z = z$ has UV soly z^n

proof suppose ① holds

the only way this can happen

$\rightarrow$ must be $B = \sum_{c_{ij} \in \text{zero}} c_{ij} A_{ij}$ some function of X_{ij}
 $(c_{13}, c_{14}, \dots, c_{1,n-1})$

ex: $B = X_{24} - X_{13} + X_{14}$

now let's look at soly

as $z \rightarrow \infty$ $B = z^m B_m(X_{ij}^{(\infty)}) + O(z^{m-1})$

ex: $B(z) = \frac{1}{X_{24}(z)} = \frac{1}{\underbrace{X_{24}(0)} + X_{24}^{(\infty)} z}$

$B(z) \rightarrow \frac{1}{X_{24}^{(\infty)}} \frac{1}{z} = z^{-1} B_{-1}(X_{24}^{\infty})$

if B has enhanced soly then

$B_m(X_{ij}^{\infty}) = 0$

$\rightarrow$ the only submatrix they satisfy are $X_{2i}^{\infty} = X_{1i}^{\infty} - X_{13}^{\infty}$

B_m satisfies the k -zero

so $B_m = \sum_{c_{ij} \in \text{zero}} c_{ij} \{ \dots \}_{ij}$

$$X_{2j}(0) = X_{1j}(0) - X_{13}(0)$$

$$X_{2j} = X_{2j}(0) + X_{2j}^{\infty} z$$



$$X_{2j}(z) = X_{1j}(z) - X_{13}(z)$$

B satisfies 1-zero then $\uparrow$ satisfy

as $z \rightarrow \infty$

$$B(X_{ij}(z)) = 0$$

$$B_m(X_{ij}^{\infty}) z^m + O(z^{m-1})$$

$$\rightarrow 0 \quad B_m(X_{ij}^{\infty}) = 0$$

B satisfies 1-zero $\Rightarrow$ enforced satisfy of $B_m \Rightarrow B_m$ satisfies 1-zero

$n=4$

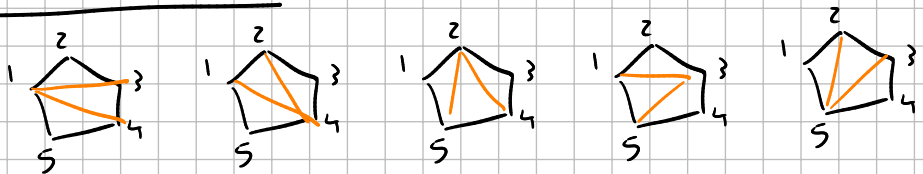
$$B = \frac{1}{X_{13}} + \frac{1}{X_{24}}$$

set

$$C_{13} = 0 \Rightarrow X_{24} = -X_{13}$$

$$B = \frac{1}{X_{13}}$$

$n=5$



Suppose

$$B = \frac{a_1}{\underbrace{X_{13} X_{14}}_{B_{-2}}} + \frac{a_2}{X_{24} X_{14}} + \frac{a_3}{X_{24} X_{25}} + \underbrace{\frac{b_1}{\underbrace{X_{13} X_{35}}_{B_{-1}}} + \frac{b_2}{\underbrace{X_{25} X_{35}}_{B_{-1}}}}_{B_{-1}}$$

$z^{-2} \quad \quad \quad z^{-1}$

require that B satisfies the 1-zero

the only term that has $\frac{1}{X_{35}}$ is $\frac{b_1}{\underbrace{X_{13} X_{35}}_{B_{-1}}} + \frac{b_2}{\underbrace{X_{25} X_{35}}_{B_{-1}}}$

$$0 = B(X^{(n)} \text{ and } X_{35}=1) = B_{-2} + \frac{b_1}{X_{13}} - \frac{b_2}{X_{13}}$$

$$0 = B(X^{(n)} \text{ and } X_{35} = \frac{1}{2}) = B_{-2} + 2 \frac{b_1}{x_{13}} - 2 \frac{b_2}{x_{13}} \quad \checkmark \quad X_{1j}$$

$$\Rightarrow B_{-2}(X^{(n)}) = 0 \quad \text{and} \quad \frac{b_1}{x_{13}} - \frac{b_2}{x_{13}} = 0 \quad \rightarrow b_2 = b_1$$

↓

$$A_4 = a \begin{array}{c} P_1 \downarrow \\ \diagup \quad \diagdown \\ 4 \quad 3 \\ \diagdown \quad \diagup \\ P_2 \end{array} + b \begin{array}{c} 1 \quad 2 \\ \diagdown \quad \diagup \\ 4 \quad 3 \end{array}$$

generic A_4 within our ansatz

$$= \frac{a}{(P_1 + P_2)^2} + \frac{b}{(P_1 + P_4)^4} = \frac{a}{X_{13}} + \frac{b}{X_{42}} = \frac{1}{X_{13}} (a - b)$$

impose zero condition: $S_{13} = 0 \Rightarrow 0 = X_{13} + X_{24} - \cancel{X_{23}}^0 - \cancel{X_{14}}^0$

if $X_{13} = -X_{24}$ then $A_4 = 0$ (assumption)

recursion relation

$$C_{ij} = X_{ij} + X_{i+1, j+1} - X_{i+1, j} - X_{i, j+1}$$

$$C_{13} = C_{14} = \dots = C_{1, n-1} = 0 \quad \Delta \Rightarrow$$

$$X_{24} = X_{14} - X_{13}$$

$$X_{25} = X_{15} - X_{13}$$

$$\vdots$$

$$X_{2n} = X_{1n} - X_{13} = -X_{13}$$

$\left. \begin{array}{l} \\ \\ \\ \end{array} \right\} n-3 \text{ equations}$

conversely, if we shift $P_2 \rightarrow P_2 + qz$
 $P_n \rightarrow P_n - qz$

the only affected X_{ij} are the ones that contain only 1 of P_2 or P_n

$$X_{1j} \quad j=3, \dots, n-1$$

$$X_{2j} \quad j=3, \dots, n$$

$$X_{1j}(z) = X_{1j}(0) + z X_{1j}^\infty$$

$$X_{1j}(z) = (P_1 + P_2 + qz + \dots + P_{j-1})^2 = \cancel{z^2 q^2}^0 + X_{1j}(0) + \cancel{2zq \cdot (P_1 + P_2 + \dots + P_{j-1})}^{X_{1j}^\infty}$$

$$X_{2j}(z) = X_{2j}(0) + \cancel{2zq \cdot (P_2 + P_3 + \dots + P_{j-1})}^{X_{2j}^\infty}$$

$$= X_{2j}(0) + X_{1j}(z) - X_{1j}(0) - \cancel{2zq \cdot P_1}^{X_{13}^\infty}$$

$$X_{13}^\infty = 2q \cdot (P_1 + P_2) = 2q \cdot P_1$$

$$X_{2i}^{\infty} = X_{1i}^{\infty} - X_{13}^{\infty}$$

Subst zeros

B is an homogeneous rational function of the X_{ij}

collect B into subterms depending on how they scale under the shift

$$P_2 \rightarrow P_2 + zq \quad P_n \rightarrow P_n - zq$$

$$B(z) = B_m + B_{m-1} + B_{m-2} + \dots$$

where B_m contains all terms that scale like z^m (as $z \rightarrow \infty$)

Prop. these are equivalent:

- B satisfies 1-zero
- each B_i satisfies 1-zero
- each B_i has enhanced UV scaling z^{i-1}

step 1 if B satisfies 1-zero then it must have enhanced scaling.

B satisfies zero news: if $X_{2i} = X_{1i} - X_{13}$ then $B = 0 \quad \forall$ other class of X_{ij}

1-zero constant
 $i=4, \dots, n$

$\rightarrow B$ must be at least linear in $c_{13}, c_{14}, c_{15}, \dots$

$$B = \sum_{i=3}^{n-1} c_{1i} \{ \dots \}_i$$

ex: $B = \frac{1}{X_{13}} + \frac{1}{X_{24}}$

$$= \frac{X_{24} + X_{13}}{X_{13} X_{24}}$$

$$0 = -c_{13} = X_{13} + X_{24} - \cancel{X_{14}} - \cancel{X_{23}}$$

What happens to B as $z \rightarrow \infty$

$$B \rightarrow z^m B_m(X_{ij}^\infty) + O(z^{m-1})$$

ex: $B = \frac{1}{x_{13}} + \frac{1}{x_{24}}$

$$x_{24}(z) = x_{24}(0) + z x_{24}^\infty$$

$$x_{13}(z) = x_{13}(0) + z x_{13}^\infty$$

$$x_{24}^\infty = -x_{13}^\infty$$

$$B = B_{-1}$$

$$B(z) = \frac{1}{x_{13}(0) + x_{13}^\infty z} + \frac{1}{x_{24}(0) - x_{13}^\infty z} = \frac{x_{13}(0) + x_{24}(0) + \cancel{x_{13}^\infty z} - \cancel{x_{13}^\infty z}}{-x_{13}^\infty z^2 + \dots} \rightarrow \frac{1}{z^2}$$

$$= \frac{1}{x_{13}^\infty z} + O(z^{-2}) \rightarrow -\frac{1}{x_{13}^\infty z} + O(z^{-2})$$

B has enhanced soling

$$\frac{1}{x_{13}} \Big|_{x_{13}=x_{13}^\infty} + O(z^{-2})$$

$$f(z) = z$$

$$g(z) = -z$$

$$f+g = 0$$

$$B = \frac{1}{1+z} + \frac{1}{2-z}$$

$$= \frac{2-z+1+z}{(1+z)(2-z)} = \frac{3}{-z^2}$$

esale series at ∞ ?

$$f(z) = \frac{1}{a+bz} \text{ expand at } \infty$$

change of variable

$$z = 1/w$$

expand at $w=0$

$$f = \frac{1}{a+b/w} = \frac{w}{wa+b} = \frac{w}{b} + O(w^2) \text{ (Taylor series)}$$

$$b = \frac{1}{bz} + O(z^{-2})$$

$$f'(0) = \frac{(wa+b) - wb}{(wa+b)^2} \Big|_{w=0} = \frac{b}{b^2} = \frac{1}{b}$$

$$\text{as } z \rightarrow \infty \quad B_m(z) \rightarrow z^m B_m(X_{ij}^\infty) + O(z^{m-1})$$

Taylor series around $z = \infty$

$$B_m \text{ has enhanced soling} \Leftrightarrow B_m(X_{ij}^\infty) = 0 \quad \forall X_{ij}^\infty$$

X_{ij}^∞ satisfies the same relations as 1-zeros

$$B_m(X_{ij}^\infty) = 0 \quad \forall X_{ij}^\infty \quad \Leftrightarrow \quad B_m \text{ satisfies 1-zero}$$

this proves that $(2) \Leftrightarrow (3)$ and $(2) \Rightarrow (1)$

suppose (1) is true

B satisfies 1-zero $\Rightarrow$ B_m satisfies 1-zero

if we let $X_{2j}(0) = X_{1j}(0) - X_{13}(0)$ then $B = 0$

then we do shift then $X_{2j}(z) = \overbrace{X_{1j}(0) + z X_{1j}^\infty}^{X_{1j}(0) + z X_{1j}^\infty} - X_{13}(z)$

$$\downarrow \\ X_{2j}(0) + z X_{2j}^\infty$$

then $B(X_{ij}(z)) \rightarrow 0$ as $z \rightarrow \infty$

since $B \geq z^m B_m(X_{ij}^\infty) + O(z^{m-1})$ it must be $B_m(X_{ij}^\infty) = 0$

let $B' = B - B_m = B_{m-1} + \dots$

ex: $B(x, y) = x - y$

$$x = 1 + z$$

$$y = -1 - z$$

$$B(1+z, -1-z) = 0$$

4 point

$$B = a \begin{array}{c} 1 \\ \swarrow \searrow \\ P_1 \quad P_2 \\ \downarrow \downarrow \\ 4 \quad 3 \end{array} + b \begin{array}{c} 1 \\ \swarrow \searrow \\ P_1 \quad P_2 \\ \downarrow \downarrow \\ 4 \quad 3 \end{array}$$

$$= \frac{a}{X_{13}} + \frac{b}{X_{24}}$$

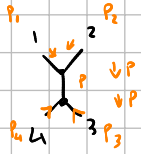
$$X_{13}, X_{24}$$

independent planar variables

$$X_{i0} = (P_i + P_{i+1} + \dots + P_{i-1})^2$$

$$X_{ii} = X_{i+i} = 0$$

$$X_{in} = 0$$



each vertex $\propto \delta$ of incoming - outgoing momenta

each internal line $= \int dp \frac{1}{p^2}$ of associated momenta

$$\int dp \underbrace{\delta(P_1 + P_2 - P)}_{f(p)} \underbrace{\delta(P_3 + P_4 + P)}_{f(p)} \frac{1}{p^2} = \frac{\delta(P_1 + P_2 + P_3 + P_4)}{(P_1 + P_2)^2} = \frac{\delta(P_1 + P_2 + P_3 + P_4)}{X_{13}}$$

$$\int dp \delta(p - q) f(p) = f(q)$$

then impose 1-zero

$$C_{13} = 0 \quad (n=4)$$

$$C_{13} = C_{14} = \dots = C_{1n-1} = 0$$

require that $C_{13} = 0 \Rightarrow B = 0 \quad \forall X$

||

$$X_{13} + X_{24} - X_{14} - X_{34}$$

(n-3) constraints

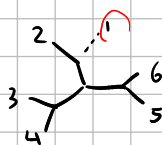
$$B = \frac{a}{X_{13}} + \frac{b}{X_{24}} = \frac{a - b}{X_{13}} \quad \text{so } \underline{a = b}$$

ex:

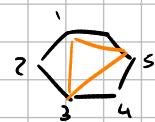
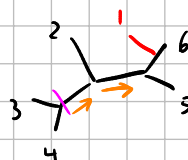
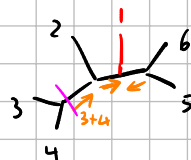
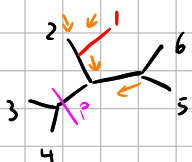
$n = 6$

choose a 6-leg diagram with leg 1 removed

start with

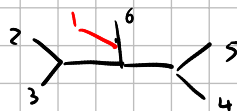
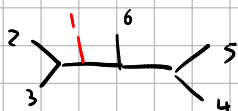
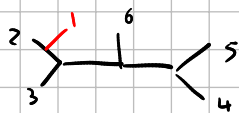
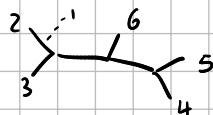


add leg 1 back in in all possible ways (according to order)



$$S_1 = \frac{a_1}{X_{13} X_{51} \underline{X_{35}}} + \frac{a_2}{X_{25} X_{15} \underline{X_{35}}} + \frac{a_3}{X_{25} X_{26} \underline{X_{35}}}$$

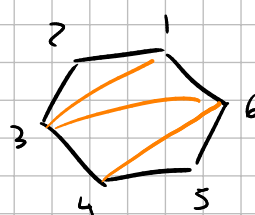
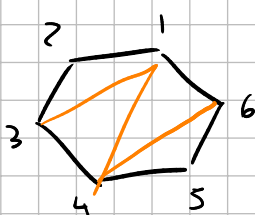
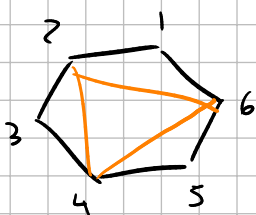
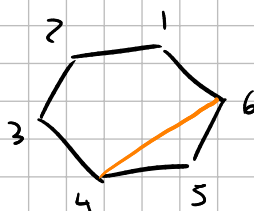
another D-subset



$$S_2 = \frac{a_4}{X_{13} X_{46} X_{41}} + \frac{a_5}{X_{24} X_{46} X_{41}} + \frac{a_6}{X_{24} X_{46} X_{26}}$$

$S_1 + S_2$ all the diagrams that contain X_{46} and X_{35}

all diagrams that have $X_{46} = (P_4 + P_5)^2$



look at $S_1 + S_2$ and see that it satisfies 1-zero ($c_{13} = c_{14} = c_{15} = 0$)

$$\Rightarrow X_{2\delta} = X_{1\delta} - X_{13} \quad \delta = 4, \dots, 6$$

if $c_{13} = c_{14} = c_{15} = 0 \Rightarrow S_1 + S_2 = 0 \quad \forall X_{i\delta}$ subject to constraint

then $c_{13} = c_{14} = c_{15} = 0 \Rightarrow S_1 = 0$ and $S_2 = 0 \quad \forall X_{i\delta}$

$$S_1 + S_2 = \frac{a_1}{X_{13} X_{51} X_{35}} + \frac{a_2}{X_{25} X_{15} X_{35}} + \frac{a_3}{X_{25} X_{26} X_{35}} + \frac{a_4}{X_{13} X_{46} X_{41}} + \frac{a_5}{X_{24} X_{46} X_{41}} + \frac{a_6}{X_{24} X_{46} X_{26}}$$

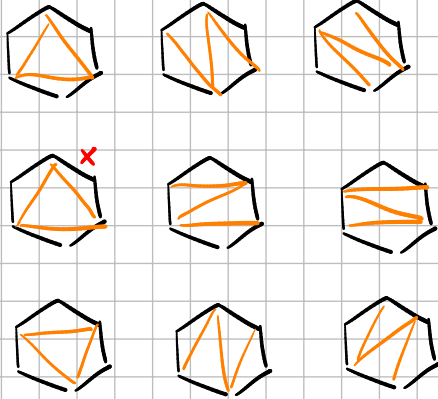
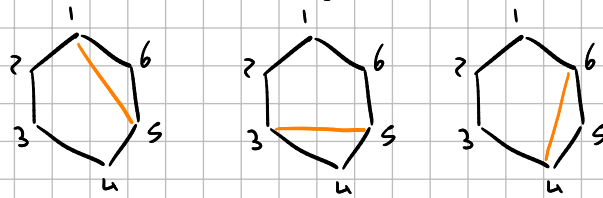
$$= \frac{1}{X_{35}} (---) + \frac{1}{X_{46}} (---)$$

$\searrow = 0 \quad \searrow = 0$

$$X_{2i} = X_{1i} - X_{1j}$$

all possible diagrams with 6 legs

group them by



splitting depending on z scaling of propagators

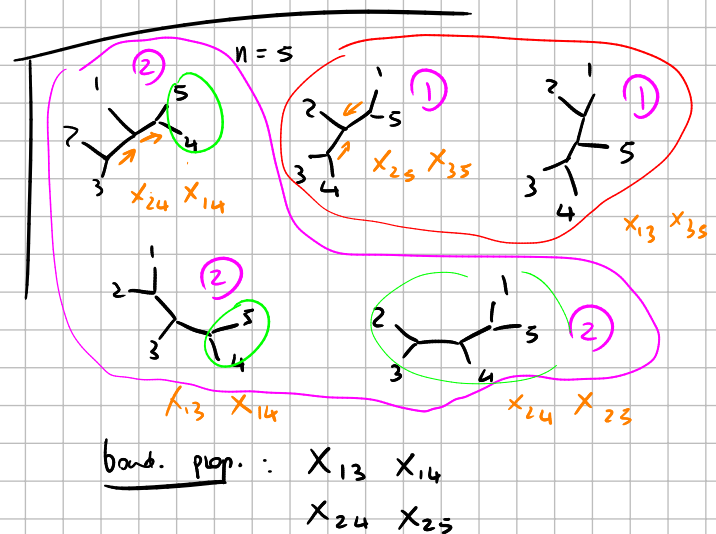
boundary propagators: the ones that scale linearly with z under BCFW shift (X_{1i} and X_{2i})

group by # of boundary propagators

→ group them by scaling under BCFW shift

$$B = \boxed{B_0} + \boxed{B_{-1}} + \boxed{B_{-2}} + \dots$$

$\downarrow$ $\downarrow$ $\downarrow$
 0 boundary 1 bound. 2 boundary
 propagators prop. prop.



1-zero and BCFW shift

up to cyclic permutations a 1-zero is the imposition of the $n-3$ constraints

$$C_{13} = C_{14} = \dots = C_{1,n-1} = 0 \quad \text{on the kinematic data.}$$

Since $C_{ij} = X_{i,i} + X_{i+1,i+1} - X_{i,i+1} - X_{i+1,i}$

the constraints can be equivalently be written as

$$\begin{cases} X_{13} + X_{24} - X_{14} - X_{23} = 0 \\ X_{14} + X_{25} - X_{15} - X_{24} = 0 \\ \vdots \\ X_{1,n-1} + X_{2n} - X_{1n} - X_{2,n-1} = 0 \end{cases} \Leftrightarrow \begin{cases} X_{24} = X_{14} - X_{13} \\ X_{25} = X_{15} - X_{13} \\ \vdots \\ X_{2n} = X_{1n} - X_{13} = -X_{13} \end{cases}$$

so the 1-zero fully fixes X_{2j} once X_{1j} is specified, and does not affect X_{ij} if $i, j \notin \{1, 2\}$.

the corresponding BCFW shift is the change $\begin{cases} p_2 \rightarrow p_2 + zq \\ p_n \rightarrow p_n - zq \end{cases}$

where z is a complex parameter and q satisfies $\begin{cases} q^2 = 0 \\ q \cdot p_2 = q \cdot p_n = 0 \end{cases}$

$\Rightarrow$ so that $(p_2 + zq)^2 = 0$
 $(p_n - zq)^2 = 0$

Let $X_{ij}(z)$ be the z -dependent planar invariant.

$X_{ij}(z)$ depends on z only if it includes exactly one of p_2, p_n in the sum, otherwise either there is no z or they cancel out.

$\rightarrow$ the only z -dependent ones are $X_{1i}(z) \quad i = 3, \dots, n-1$
and $X_{2j}(z) \quad j = 4, \dots, n$.

Let $\tilde{X}_{ij}$ denote the coefficient of the largest power of z appearing in $X_{ij}(z)$.

$$X_{1j}(z) = (p_1 + p_2 + zq + \dots + p_{j-1})^2 = (p_1 + p_2 + \dots + p_{j-1})^2 + (zq)^2 + 2zq \cdot (p_1 + p_2 + \dots + p_{j-1})$$

$$= X_{1j}(0) + z \cdot 2q \cdot (p_1 + p_2 + \dots + p_{j-1}) \Rightarrow X_{1j}^\infty = 2q \cdot (p_1 + p_2 + \dots + p_{j-1})$$

($j = 3, \dots, n-1$)

$$X_{2j}(z) = (p_2 + zq + \dots + p_{j-1})^2 = \dots$$

$$= X_{2j}(0) + z \cdot 2q \cdot (p_2 + p_3 + \dots + p_{j-1}) \Rightarrow X_{2j}^\infty = 2q \cdot (p_2 + p_3 + \dots + p_{j-1})$$

($j = 4, \dots, n$)

in all other cases $X_{ij}(z) = X_{ij}(0)$ so $X_{ij}^\infty = X_{ij}(0)$

note that $X_{2j}^\infty = X_{1j}^\infty - 2q \cdot p_1 = X_{1j}^\infty - X_{13}^\infty$ ($j = 4, \dots, n$)

so the X_{ij}^∞ satisfy the same constraint as the 1-zero.

Subset zeros

if $\lambda \neq 0$ $B(\lambda X_{ij}) = \lambda^s B(X_{ij})$ for some $s \in \mathbb{Z}$

Let $B(X_{ij})$ be an homogeneous rational function of the X_{ij} variables.

We will write $B(X) = B_m(X) + B_{m-1}(X) + \dots$

where $B_i(X)$ is the sum of all the terms of B that scales like z^m as $z \rightarrow \infty$ under the BCFW shift. (There are finitely many B_i)

example: $n=5$, independent X_{ij} are $X_{13}, X_{14}, X_{24}, X_{25}, X_{35}$

$\downarrow$ scale like z $\downarrow$ scales like z^0

if $B = \frac{X_{13}}{X_{24} X_{14}} + \frac{X_{35}}{X_{13} X_{24}} + \frac{1}{X_{35}}$ (note it's homogeneous)

$\downarrow$ scale like $\frac{z}{z^2} = \frac{1}{z}$ $\downarrow$ scale like $\frac{1}{z^2}$ $\downarrow$ scale like z^0

then $B_0 = \frac{1}{X_{35}}$ $B_{-1} = \frac{X_{13}}{X_{24} X_{14}}$ $B_{-2} = \frac{X_{35}}{X_{13} X_{24}}$ and $B = B_0 + B_{-1} + B_{-2}$

Def: we say a function $B = B(X_{ij})$ satisfies the 1-zero if

$$X_{2j} = X_{1j} - X_{13} \Rightarrow B(X) = 0 \quad (\forall \text{ choice of } X_{ij} \text{ with } i,j \neq 2)$$

45 mins

10:25

Prop. The following are equivalent.

- ① B satisfies the 1-zero
- ② each B_i satisfies the 1-zero

Proof ② $\Rightarrow$ ① is trivial.

Suppose ① is true. and let $B(z) = B(X_{ij}(z))$

note that as $z \rightarrow \infty$ $B(z) = z^m B_m(X_{ij}^\infty) + o(z^{m-1})$

This is because B is a rational function and m is the largest power of z

appearing. ex: if $B = \underbrace{\frac{X_{13}}{X_{24} X_{14}}}_{B_{-1}} + \underbrace{\frac{X_{35}}{X_{13} X_{24}}}_{B_{-2}}$

then $B(z) = B_{-1}(z) + o(z^{-2})$

$$= \frac{X_{13}(n) + z X_{13}^\infty}{(X_{24}(n) + z X_{24}^\infty)(X_{14}(n) + z X_{14}^\infty)} + o(z^{-2})$$

$$= \frac{1}{z} \frac{X_{13}^\infty}{X_{24}^\infty X_{14}^\infty} + o(z^{-2}) = z^{-1} B_{-1}(X_{ij}^\infty) + o(z^{-2})$$

Since $X_{2j}^\infty = X_{1j}^\infty - X_{13}^\infty$

note that if $X_{2j}(n) = X_{1j}(n) - X_{13}(n)$ then $X_{2j}(z) = X_{1j}(z) - X_{13}(z) \forall z$

so $B(z)$ satisfies the zero $\forall z$. But then

$$0 = z^m B_m(X_{ij}^\infty) + o(z^{m-1}) \quad \forall z \quad \text{so it must be } B_m(X_{ij}^\infty) = 0$$

the only constraints on the X_{ij}^∞ are the same as the 1-zero,

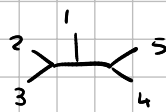
so this means B_m satisfies the 1-zero.

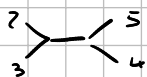
Then we proceed by induction. if B satisfies the 1-zero and B_m satisfies the 1-zero then $B' = B - B_m = B_{m-1} + B_{m-2} + \dots$ satisfies the 1-zero. Following the same steps as above we then prove that B_{m-1} satisfies the 1-zero and so on.

Generic amplitude: arbitrary linear combination of all ordered Feynman diagrams for n external legs.

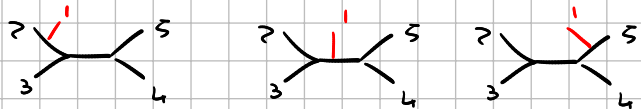
D-subsets

A D-subset at n -point is the subset of the generic amplitude that contains all diagrams obtained by starting with a particular Feynman diagram, removing leg 1, then reattaching it in all possible ways (within the order).

for example: $n=5$, start with 

• remove leg 1: 

• add back leg 1 in all ways:

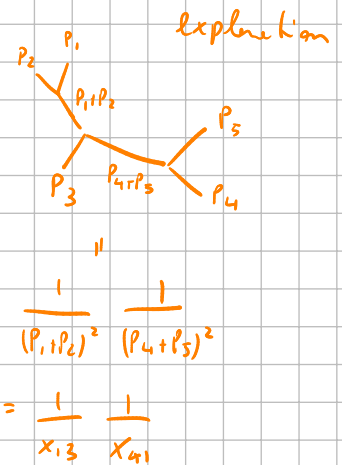


the associated D-subset is

$$a_1 \text{ (diagram 1)} + a_2 \text{ (diagram 2)} + a_3 \text{ (diagram 3)}$$

$$= \frac{a_1}{X_{13} X_{41}} + \frac{a_2}{X_{24} X_{41}} + \frac{a_3}{X_{24} X_{52}}$$

explanation



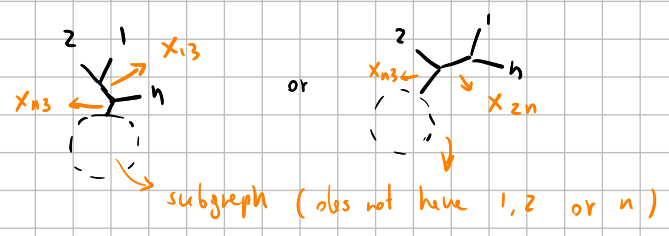
$$\frac{1}{(p_1 + p_2)^2} \frac{1}{(p_4 + p_5)^2}$$

$$= \frac{1}{X_{13}} \frac{1}{X_{41}}$$

Def: A "boundary propagator" is an X_{ij} that depends on z under a BCFW shift (so $X_{13}, \dots, X_{1n-1}$ and $X_{24}, \dots, X_{2n}$)

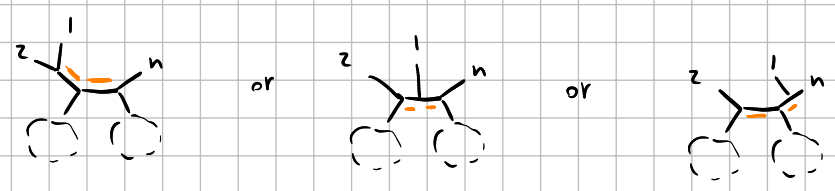
Now, if B is our generic amplitude at n points, we split it in groups depending on the number of boundary propagators each term has. This is the same as splitting $B = B_m + B_{m-1} + \dots$ from earlier, since a term with k boundary propagators scales as z^{-k} when $z \rightarrow \infty$.

A graph with 1 boundary propagator looks like



$$\frac{1}{x_{12}} = \frac{1}{x_{12}} \quad n=4 \quad (\text{only } 1)$$

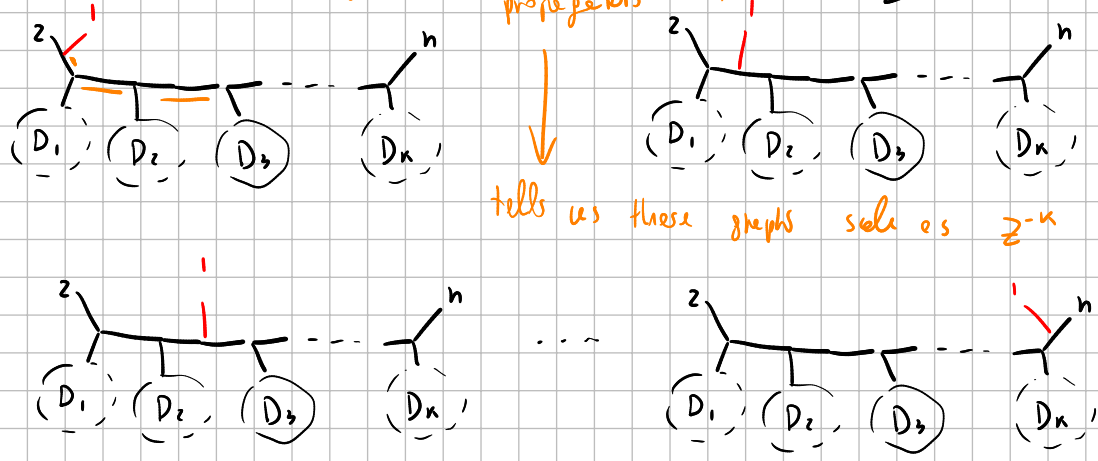
2 boundary operators



$$\frac{1}{x_{13} x_{35}} = \frac{1}{x_{25} x_{35}} \quad n=5$$

k boundary operators:

orange lines count # boundary propagators (k)

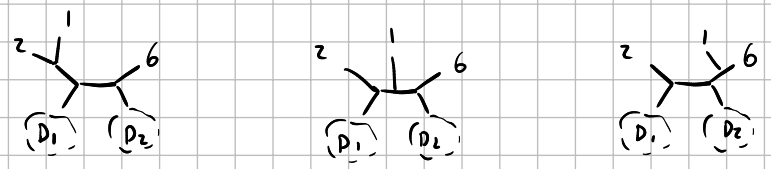


It should be clear from the image that removing leg 1 and re-adding it does not change the # of boundary propagators, so we can partition each B_n in D -subsets (by going through all choices of subgraphs $D_1, \dots, D_n$).

Let's see an example:

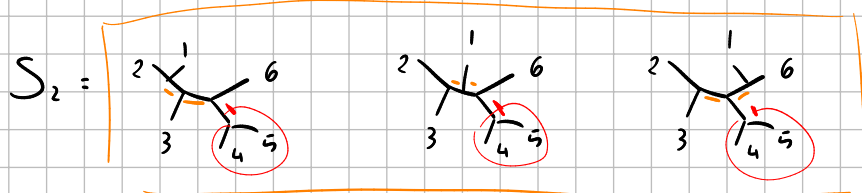
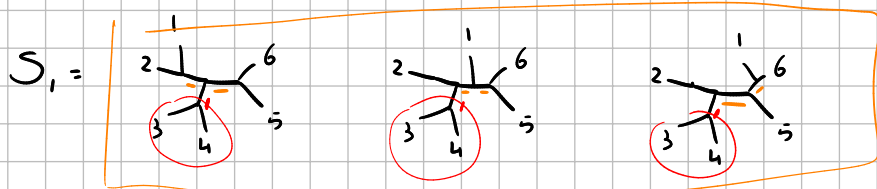
$n=6, \quad 2 \text{ boundary propagators.}$

must look like



the choices for D_1, D_2 are: $(D_1 = \frac{1}{3} \frac{1}{4}, D_2 = \frac{1}{5})$ and $(D_1 = \frac{1}{3} \frac{1}{4}, D_2 = \frac{1}{4} \frac{1}{5})$

so there are 2 D-subsets:



Since $S_1 + S_2 = B_{-2}$ we know it must satisfy 1-zero if B does.

However, so do S_1 and S_2 independently, since

S_1 has a factor of X_{35} coming from the piece

and S_2 has a factor of X_{46} coming from the piece.

$$\Rightarrow S_1 + S_2 = \frac{1}{X_{35}}(\dots) + \frac{1}{X_{45}}(\dots) = 0 \quad \forall X_{35}, X_{45} \text{ only if}$$

the coefficients of $\frac{1}{X_{35}}$ and $\frac{1}{X_{45}}$ vanish independently $\Rightarrow S_1 = 0, S_2 = 0$.

Moreover, imposing the 1-zero uniquely determines S_1 and S_2 up to an overall coefficient. Note that

$$S_1 = \frac{1}{X_{35}} \left(\frac{a_1}{X_{13} X_{15}} + \frac{a_2}{X_{25} X_{15}} + \frac{a_3}{X_{25} X_{26}} \right) = 0$$

$$\Leftrightarrow \frac{a_1}{X_{13} X_{15}} + \frac{a_2}{X_{25} X_{15}} + \frac{a_3}{X_{25} X_{26}} = 0$$

using the fact that the 1-zero constrains $X_{26} = -X_{13}$

we get $0 = \frac{a_1}{X_{13} X_{15}} + \frac{a_2}{X_{25} X_{15}} - \frac{a_3}{X_{25} X_{13}}$

$$= \frac{a_1 X_{25} + a_2 X_{13} - a_3 X_{15}}{X_{13} X_{15} X_{25}} = 0$$

so it must be $a_1 X_{25} + a_2 X_{13} - a_3 X_{15} = 0$

$$X_{25} = X_{15} - X_{13}$$

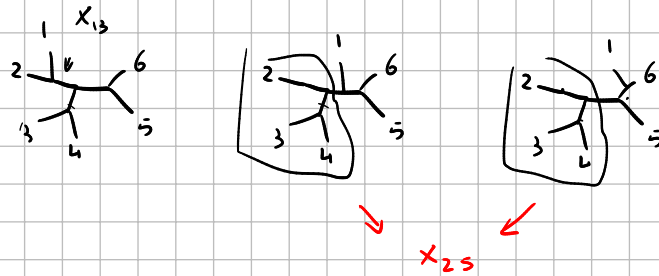
$$= a_1 (X_{15} - X_{13}) + a_2 X_{13} - a_3 X_{15} = 0$$

$$= X_{15} (a_1 - a_3) + X_{13} (a_2 - a_1)$$

Since X_{13}, X_{15} are arbitrary it must be $a_1 = a_2$ and $a_1 = a_3$

$$\text{so } a_1 = a_2 = a_3.$$

alternative



X_{2j} are arbitrary and X_{1i} fixed by X_{2j}

$$\begin{cases} X_{2n} = -X_{13} \\ X_{2j} = X_{1j} - X_{13} \end{cases}$$

$$X_{1j} = X_{2j} + X_{26}$$

since X_{25} arbitrary

$$S_1 = \left(\begin{array}{c} \text{---} \\ \text{---} \end{array} \right)_1 + \frac{1}{X_{25}} \left(\begin{array}{c} \text{---} \\ \text{---} \end{array} \right)_2 = 0 \Rightarrow \begin{cases} \left(\begin{array}{c} \text{---} \\ \text{---} \end{array} \right)_1 = 0 \\ \left(\begin{array}{c} \text{---} \\ \text{---} \end{array} \right)_2 = 0 \end{cases}$$

eq 22 or Redline

inlec :

$$\begin{cases} a + b = 0 \\ a + 2b = 0 \end{cases} \Rightarrow \begin{cases} a = 0 \\ b = 0 \end{cases}$$

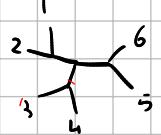
easy that

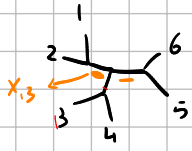
$$a_2 \left(\begin{array}{c} \text{---} \\ \text{---} \end{array} \right)_{X_{25}} + a_3 \left(\begin{array}{c} \text{---} \\ \text{---} \end{array} \right)_{X_{25}} = 0$$

$$\frac{1}{X_{25}} \left(\frac{a_2}{X_{15}} + \frac{a_3}{X_{26}} \right) = 0$$

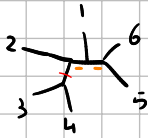
$$X_{26} = X_{16} - X_{13}$$

looks like eq. 22 is wrong (which is good!)

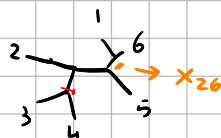
(otherwise the coefficient of  would be zero)



$$\frac{1}{X_{13} X_{15}}$$



$$\frac{1}{X_{25} X_{15}}$$



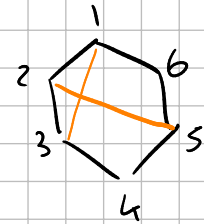
$$\frac{1}{X_{25} X_{26}}$$

$$X_{26} = -X_{13}$$

$$-\frac{1}{X_{25} X_{13}}$$



this does not correspond to
a Feynman diagram



↑
interweaving
chords,
not a
triangulation

$$S_1 = \alpha_1 \text{ (graph 1) } + \alpha_2 \text{ (graph 2) } + \alpha_3 \text{ (graph 3) }$$

$$= \frac{1}{X_{35}} \left(\alpha_1 \text{ (graph 1 with P) } + \alpha_2 \text{ (graph 2 with P) } + \alpha_3 \text{ (graph 3 with P) } \right)$$

all graphs with $n=5$ that have 2 boundary propagator